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Key Points:
- New paleomagnetic data from the Amuria Block (AMU) provide the straightforward constraints on the evolution of Mongol-Okhotsk suture (MOS)
- New data demonstrate that AMU and North China Block remained the tectonically coherence during Late Jurassic-Early Cretaceous
- We provide a reconstruction of Siberia, AMU, and NCB and propose a new subduction-collision tectonic model of the MOS at some key times

Supporting Information:
- Supporting Information S1

Correspondence to:
S. Zhang,
shzhang@cugb.edu.cn

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New Late Jurassic to Early Cretaceous Paleomagnetic Results From North China and Southern Mongolia and Their Implications for the Evolution of the Mongol-Okhotsk Suture

Qiang Ren[1], Shihong Zhang[1], Yuqi Wu[1], Tianshui Yang[1], Yangjun Gao[1], Sukhbaatar Turbold[2,3], Hanqing Zhao[1], Huaichun Wu[1], Haiyan Li[1], Hairuo Fu[1], Bei Xu[4], Jinjiang Zhang[4], and Onongoo Tomurtogoo[2]

[1]State Key Laboratory of Biogeology and Environment Geology, China University of Geosciences, Beijing, China, [2]Institute of Paleontology and Geology Mongolian Academy of Science, Ulaanbaatar, Mongolia, [3]Institute of Petrology and Structural Geology, Charles University, Prague, Czech Republic, [4]The Key Laboratory of Orogenic Belts and Crustal Evolution, Ministry of Education, School of Earth and Space Sciences, Peking University, Beijing, China

Abstract To better constrain the evolution of the Mongol-Okhotsk suture, we carried out new paleomagnetic studies on Sharilyn Formation (~155 Ma) and Tsagantsav Formation (~130 Ma) in southern Mongolia, Amuria Block (AMU), and Tuchengzi Formation (~140 Ma) and Dadianzi/Yixian Formation (~130 Ma) in the Yanshan belt, North China Block (NCB). A total of 719 collected samples (from 100 sites) were subjected to stepwise thermal demagnetization. After a low-temperature component of viscous magnetic remanence acquired in the recent field was removed, the stable high-temperature components were isolated from most samples. The high-temperature components from each rock unit passed a fold test and a reversal test, indicating their primary origins. The corresponding paleomagnetic poles were thus calculated. For AMU, the ~155 Ma pole is at 74.7°N/232.5°E (A_{95} = 3.7°), the ~130 Ma pole at 74.6°N/194.7°E (A_{95} = 2.9°); for the NCB, the ~140 Ma pole is at 82.7°N/208.6°E (A_{95} = 4.3°), the ~130 Ma pole at 80.5°N/197.4°E (A_{95} = 2.3°). By combining our new results with the published data, we refined the 155–100 Ma segment of the apparent polar wander paths for AMU and NCB, which can demonstrate that these two blocks have been tectonically coherent (AMU-NCB) during 155–100 Ma. Comparison of the apparent polar wander paths, however, revealed a latitudinal plate convergence of 14.3° ± 6.9° and ~19.0° relative rotation between Siberia and the AMU-NCB after ~155 Ma. Large-scale latitudinal convergence likely ceased by ~130 Ma, although some relative rotation between them continued along the Mongol-Okhotsk suture until ~100 Ma.

1. Introduction

The Mongol-Okhotsk suture (MOS) is widely accepted as an important tectonic boundary between Siberia and the Amuria (AMU)-North China Block (NCB; Figure 1a), and thus bears important information to understand the amalgamation of the eastern part of the Eurasian continent (Cogné et al., 2005; Enkin et al., 1992; Halim et al., 1998; Kravchinsky, Cogné, et al., 2002; Metelkin et al., 2010; Ren et al., 2016; Tang et al., 2015; Tomurtogoo et al., 2005; Van der Voo et al., 2015; Wu, Kravchinsky, Gu, et al., 2017; Wu, Kravchinsky, & Potter, 2017; Zonenshain et al., 1990; Zorin, 1999). Regarding this issue, paleomagnetism remains the most powerful tool for studying plate motion and has provided a great deal of independent evidence for paleogeographic reconstruction. The current kinematic models supported by paleomagnetic data are mainly based on comparisons between the paleomagnetic databases from each side of the MOS (e.g., Cogné et al., 2005; Enkin et al., 1992; Gilder & Courtillot, 1997; Halim et al., 1998; Kravchinsky, Cogné, et al., 2002; Metelkin et al., 2010; Ren et al., 2016; Van der Voo et al., 2015; Wu, Kravchinsky, Gu, et al., 2017; Wu, Kravchinsky, & Potter, 2017; Zhao et al., 1990). However, although the apparent polar wander path (APWP) of the Siberia craton on the north side of the MOS has been updated (Metelkin et al., 2010, 2012), that of the south side still has problems. The south side of the MOS contains several tectonic units, such as southern Mongolia, northeastern China, the Inner Mongolia belt, and the Yanshan belt of the North China craton. Some areas have been intensively deformed since the Late Jurassic by episodic folding, thrusting, and regional extension, whereas others have been fairly stable (Cogné et al., 2005; Davis et al., 2001; Ren et al., 2016; Y. Wang et al., 2018). Most paleomagnetic data from the south side of the MOS are from the North

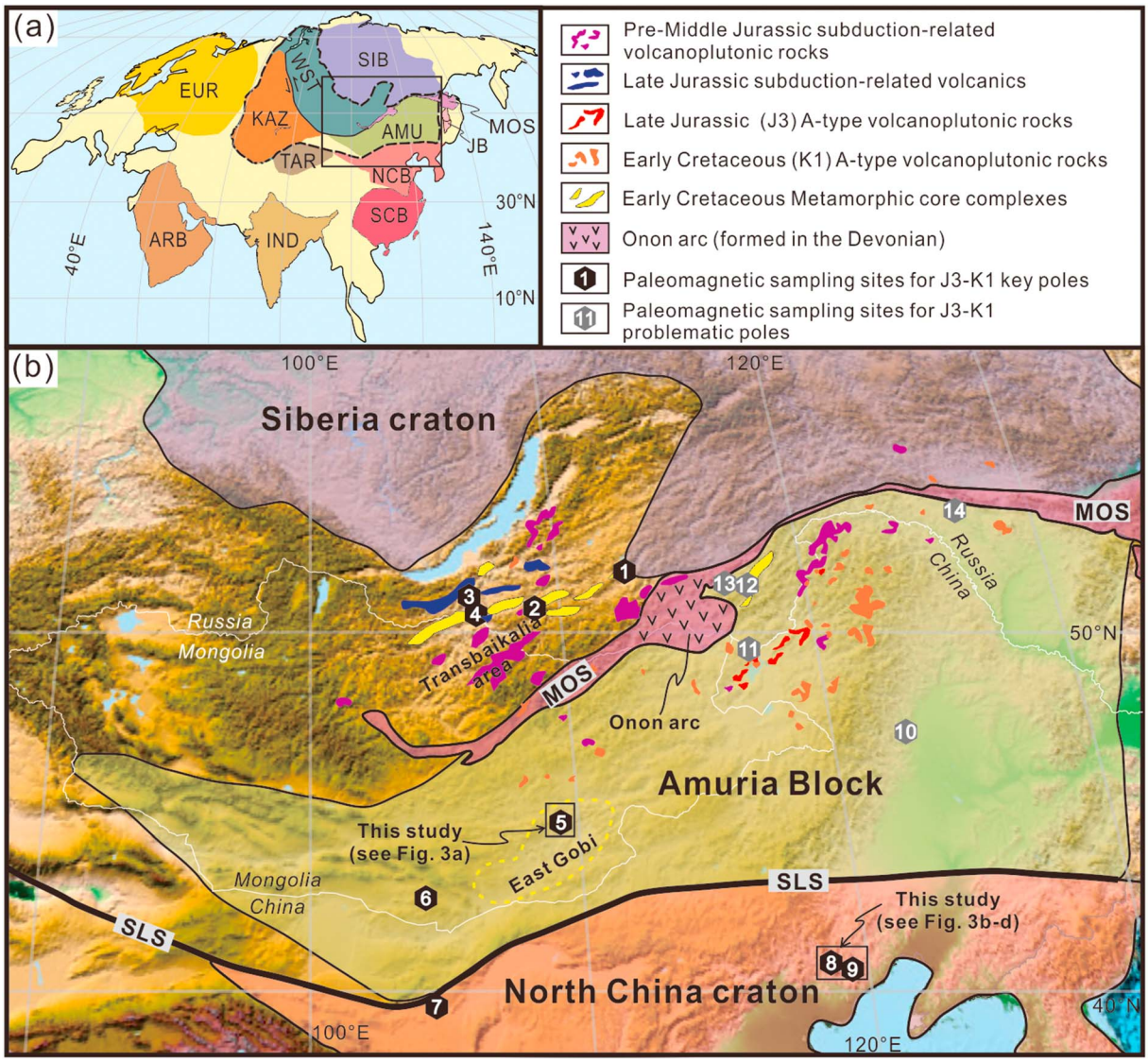


Figure 1. (a) Tectonic map of Eurasia. (b) Topographic and tectonic map of MOS between the Siberia Craton in the north, and the AMU-NCB in the South. The outlines of Siberia, AMU, and NCB modified after the Domeier and Torsvik (2014) and Van der Voo et al. (2015). The number of the paleomagnetic sites corresponded to the poles in Table 2. The Late Jurassic subduction-related volcanics in transbaikalia area, the Late Jurassic and Early Cretaceous A-type volcanoplutonic rocks modified from Donskaya et al. (2013), W. L. Xu et al. (2013), and T. Wang et al. (2015), respectively. The Early Cretaceous Metamorphic core complexes came from Donskaya et al. (2008). Background image created using Gplates. The dotted lines delineate the outlines of the Central Asian Orogenic Belt in (a). The abbreviations: EUR = Europe; KAZ = Kazakhstan Block; SIB = Siberia Craton; AMU = Amuria Block; NCB = North China Block; SCB = South China Block; JB = Jiamusi-Bureya Block; WST = West Siberia basin and Sayan-Tuwa Block; TAR = Tarim Block; IND = India Block; ARB = Arabia Block; MOS = Mongol-Okhotsk suture; SXCY = Solonker-Xra Moron-Changchun-Yanji suture.

China craton (Figure 1b; e.g., Gilder & Courtillot, 1997; Ren et al., 2016; Van der Voo et al., 2015; Wu, Kravchinsky, Gu, et al., 2017), and would benefit from more precise age constraints, wider spatial comparison and averaging, and recent advances in paleomagnetic technologies, such as testing for inclination shallowing (Van der Voo et al., 2015; Wu, Kravchinsky, Gu, et al., 2017; Wu, Kravchinsky, & Potter, 2017) and secular variations (Ren et al., 2016). Therefore, our new investigations were extended into the remote desert of southern Mongolia, part of the AMU, focused on well-dated successions, and added data from both volcanic and clastic rocks.

In this paper, we report new paleomagnetic results from the ~155 Ma Sharilyn Formation (Fm) sandstones and ~130 Ma Tsagantsav Fm basaltic lavas of southern Mongolia, AMU, and the ~140 Ma Tuchengzi Fm sandstones and ~130 Ma Dadianzi/Yixian Fm sedimentary rocks and basaltic lavas of the Yanshan belt,

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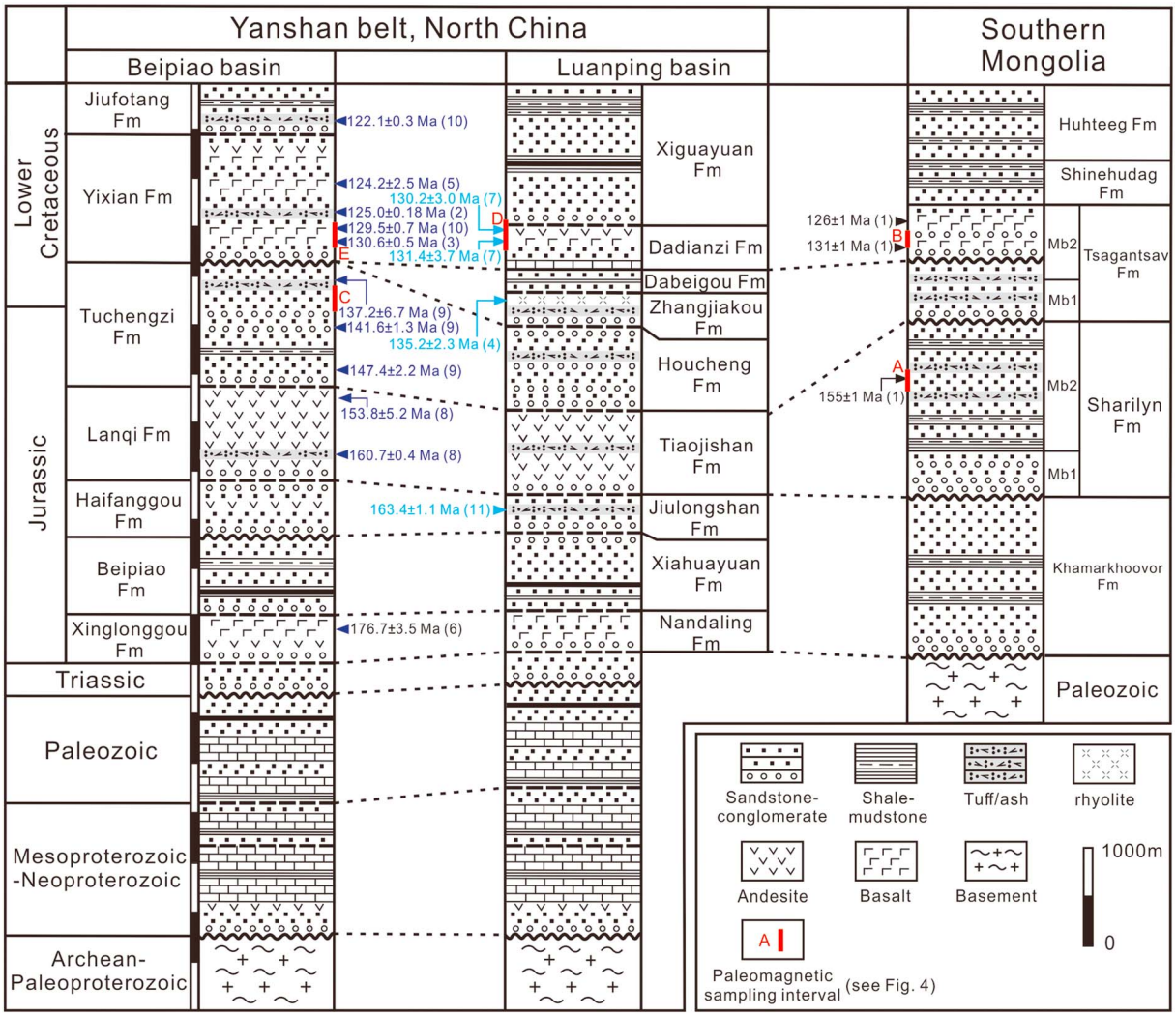


Figure 2. Stratigraphic sequence of the Yanshan belt, North China and Southern Mongolia (modified after Graham et al., 2001; H. Xu et al., 2017; and H. Zhang, Guo, et al., 2008). The references for age: (1) Graham et al. (2001), (2) Swisher et al. (2002), (3) Peng et al. (2003), (4) H. Zhang et al. (2005), (5) Zhu et al. (2007), (6) W. Yang and Li (2008), (7) H. Zhang, Guo, et al. (2008), (8) H. Zhang, Wang, et al. (2008), (9) H. Zhang, Wei, et al. (2008), (10) Chang et al. (2009), and (11) H. Y. Chen et al. (2014).

NCB. Our new data for the AMU and NCB, together with data from Siberia and geological and tomographic information, provide insights into the paleogeographic relationships of the three blocks and impose important constraints on the subduction-collision tectonic model of the MOS at some key points in geologic time.

2. Geological Setting and Sampling

2.1. Southern Mongolia, Amuria

The AMU occupies the eastern part of the Central Asian Orogenic Belt between the Siberia craton and the NCB (Figure 1a), bounded by two sutures: the Solonker suture to the south and the MOS to the north (Figure 1b; Cogné et al., 2005; Domeier & Torsvik, 2014; Kravchinsky, Cogné, et al., 2002; Y. T. Yang et al., 2015; Zonenshain et al., 1990). It consists of a number of Precambrian massifs or terranes (e.g., Khingan, central and southeastern Mongolia, and Argun), which had amalgamated as a large, united tectonic block by the Late Paleozoic (Cogné et al., 2005; Kravchinsky, Cogné, et al., 2002; Zonenshain et al., 1990).

We carried out new paleomagnetic investigations in the East Gobi Basin of southern Mongolia, AMU (Figure 1b). In this region, Paleozoic volcanic arc-related marine siliciclastics and carbonates were metamorphosed across much of the area and formed the basement (Figure 2; Graham et al., 2001; Lamb & Badarch,

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1997; Prost, 2004; Traynor & Sladen, 1995). This basement is unconformably overlain by Jurassic to Tertiary volcanic and clastic strata (Figure 2) that were deposited in separate, commonly fault-bounded basins (Graham et al., 2001; Johnson, 2004; Meyerhoff & Meyer, 1987a, 1987b; Prost, 2004). The Upper Jurassic–Lower Cretaceous volcanic and sedimentary rocks are widely exposed in the East Gobi.

2.1.1. The Upper Jurassic Sharilyn Fm

Unmetamorphosed Upper Jurassic Sharilyn Fm sedimentary rocks are mainly preserved in the northwest of the Har Hotol area and are composed of red sandstones with some shales, mudstones and tuff interbeds (Member 2), and sandstone-conglomerates (Member 1; Figure 2; Graham et al., 2001; Johnson, 2004). This formation is disconformably overlain by the Lower Cretaceous Tsagantsav Fm, and disconformably underlain by the Lower–Middle Jurassic Khamarkhoovor Fm (Figure 2; Graham et al., 2001; Jerzykiewicz & Russell, 1991). The thickness of the Sharilyn Fm is approximately 1,300 m, and its fresh sandstones are amenable to paleomagnetic study. In this area (Figure 3a), a tuff bed within Member 2 of the Sharilyn Fm has been dated at 155 ± 1 Ma based on the $^{40}Ar/^{39}Ar$ method (Graham et al., 2001).

A total of 124 individual oriented sandstone samples from 16 sites were collected from the Sharilyn Fm near the Har Hotol area (Figure 3a). Of these samples, 42 samples from five sites were collected from section I in 2014, and 82 samples from 11 sites were drilled from section II in 2015. The stratigraphic thickness between the two sections is ~80 m, and the strata of section I lie on the section II (Figure 4, Column A).

2.1.2. The Lower Cretaceous Tsagantsav Fm

The Lower Cretaceous volcanic and clastic rocks of the Tsagantsav Fm are widely exposed in the Har Hotol area and are newly divided into two members, separated by an unconformity (Figure 2; Graham et al., 2001; Johnson, 2004): (1) Member 1 consists of sandstones and conglomerates with some mudstone and tuff interbeds and is approximately 600 m thick; (2) the ~300-m-thick strata of Member 2 are composed of basaltic lava and sandstone-conglomerates with some mudstone and tuff interbeds, and disconformably overlie Member 1. The basalt flows at the bottom of Member 1 of the Tsagantsav Fm yielded a plagioclase $^{40}Ar/^{39}Ar$ age of 131 ± 1 Ma (Graham et al., 2001).

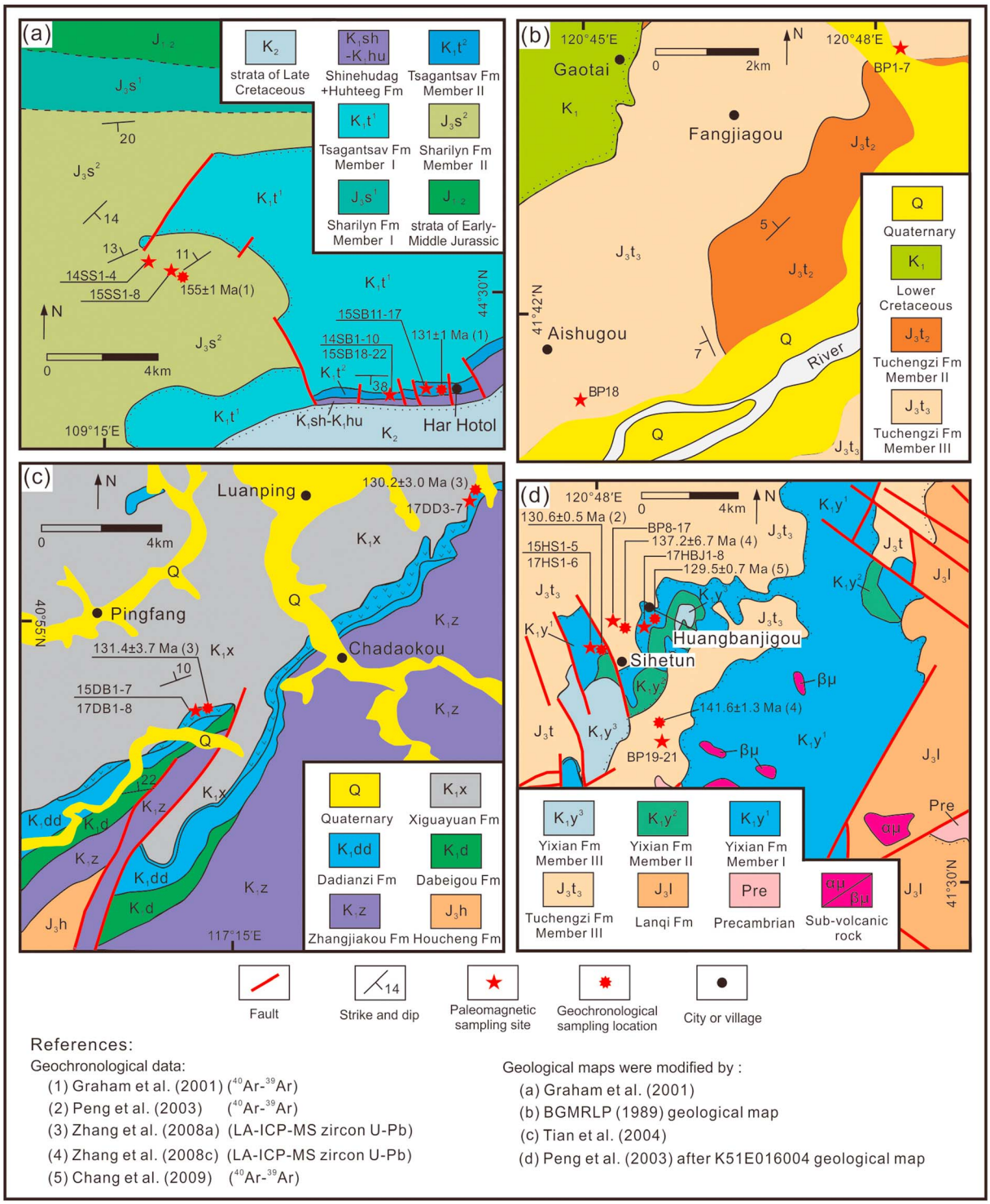
In this area, we collected 218 samples from 28 sites from the dark gray basalts of the lower part of the Tsagantsav Fm (~130 Ma), including two sections that are ~1 km apart (Figure 3a). The strata of section II (59 samples from 8 sites) overlie the strata of section I (159 samples from 20 sites; Figure 4, Column B). The attitudes of the basaltic lava were measured from the red beds and crystal tuff interbeds (Figures 5b and 5c).

2.2. Yanshan Belt, North China

The Yanshan belt, a crustal deformation belt at the northern margin of the NCB, has been tectonically coherent with respect to the interior of the North China craton since the Late Jurassic (Ren et al., 2016). It has Archean to Paleoproterozoic metamorphic rocks as the stable craton basement, covered by thick Mesoproterozoic to Paleozoic sedimentary successions. Those successions, in turn, are unconformably overlain by Mesozoic and Cenozoic terrestrial volcanic and clastic strata (Figure 2; P. J. Chen, 1999; Davis et al., 2001; Ren et al., 2016; Y. Wang et al., 2013; J. F. Zhang, 2002).

2.2.1. The Upper Jurassic to Lower Cretaceous Tuchengzi Fm

The 400–2,000-m-thick Tuchengzi Fm is widely exposed in western Liaoning Province in the Yanshan belt. It is divided into three members and consists mainly of purplish-red siltstones, sandstones, and conglomerates with intercalated tuff in the lower member (Member 1), gray-purple conglomerates intercalated with sandstones and siltstones in the middle member (Member 2), and green cross-bedded sandstone interbedded with green tuffaceous sandstones and conglomerates in the upper member (Member 3; Bureau of Geology and Mineral Resources of Liaoning Province, 1989; P. J. Chen et al., 2006). This formation is conformably or disconformably underlain by the Tiaojishan Fm and is conformably overlain by the Yixian Fm (Figure 2). Over recent decades, several U–Pb and $^{40}Ar/^{39}Ar$ ages of the Tuchengzi Fm tuffs of the Beipiao Basin of western Liaoning have been reported, which indicated the duration of 147.4–137.2 Ma, crossing the Jurassic-Cretaceous boundary (Figure 2; Chang et al., 2009; Swisher et al., 2002; H. Xu et al., 2012; H. Zhang, Wang, et al., 2008). We collected 136 samples from 21 sites from Member 3 of the Tuchengzi Fm near the well-dated (U–Pb zircon, 141.6 ± 1.3 and 137.2 ± 6.2 Ma) tuff successions reported by H. Zhang, Wang, et al. (2008; Figures 3b, 3d, and 4, Column C).



References:

Geochronological data:

(1) Graham et al. (2001) (^{40}Ar-^{39}Ar)
(2) Peng et al. (2003) (^{40}Ar-^{39}Ar)
(3) Zhang et al. (2008a) (LA-ICP-MS zircon U-Pb)
(4) Zhang et al. (2008c) (LA-ICP-MS zircon U-Pb)
(5) Chang et al. (2009) (^{40}Ar-^{39}Ar)

Geological maps were modified by :

(a) Graham et al. (2001)
(b) BGMRLP (1989) geological map
(c) Tian et al. (2004)
(d) Peng et al. (2003) after K51E016004 geological map

Figure 3. Simplified geological map for sampling sections: (a) Har Hotol of southern Mongolia, (b) south Beipiao city, (c) south Luanping city, and (d) Sihetun, Beipiao.

2.2.2. The Lower Cretaceous Dadianzi and Yixian Fms

The Dadianzi Fm, which unconformably underlies the Xiguayuan Fm and conformably overlies the Dabeigou Fm (Figure 2), is mainly exposed in the Luanping Basin and consists of volcanic and lacustrine sedimentary rocks (Liu et al., 2002; Ohta et al., 2011; H. Zhang, Guo, et al., 2008). Two fossil assemblages can be clearly identified from the fossil-bearing formation (the lower one is the *Yanshanina-Cypridea-Rhinocypris* assemblage

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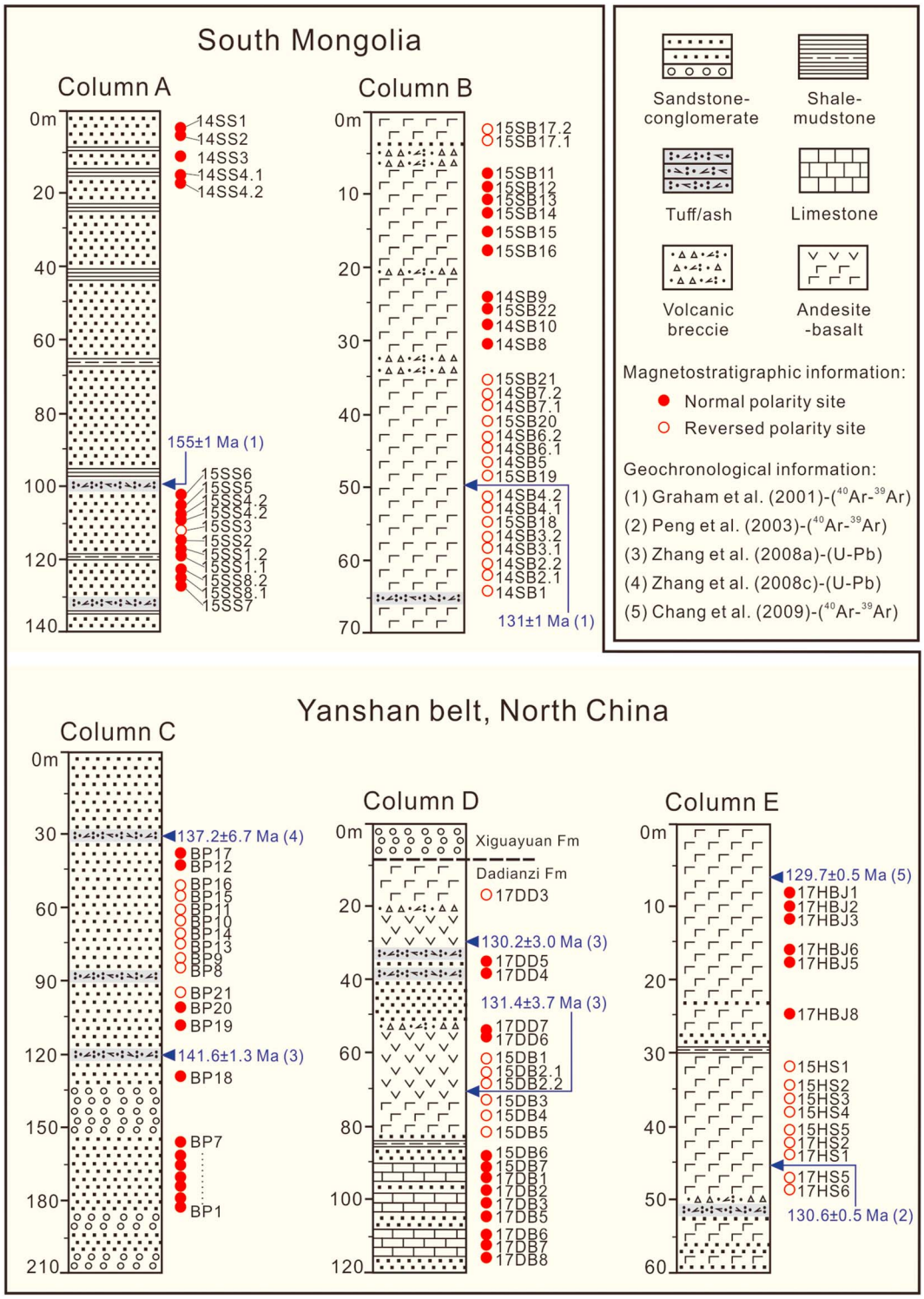


Figure 4. Lithologic strata for the paleomagnetic sampling sections and magnetostratigraphic information. The columns see Figure 3: Column A = Sharilyn Fm; Column B = Tsagantsav Fm; Column C = Tuchengzi Fm; Column D = Dadianzi Fm; and Column E = Yixian Fm.

and upper one is *Cypridea-Yanshanina-Timiriasevia* assemblage), and belong to the Early Cretaceous (Liu et al., 2002; Pang et al., 2002). At the top of the Dadianzi Fm is a special bed of basaltic andesite lava flows with a thickness of about 50 to 100 m, which has been dated at 131.4 ± 3.7 and 130.2 ± 3.0 Ma by the zircon U–Pb LA-ICP-MS method (Figure 2; H. Zhang, Guo, et al., 2008). Corresponding to these two

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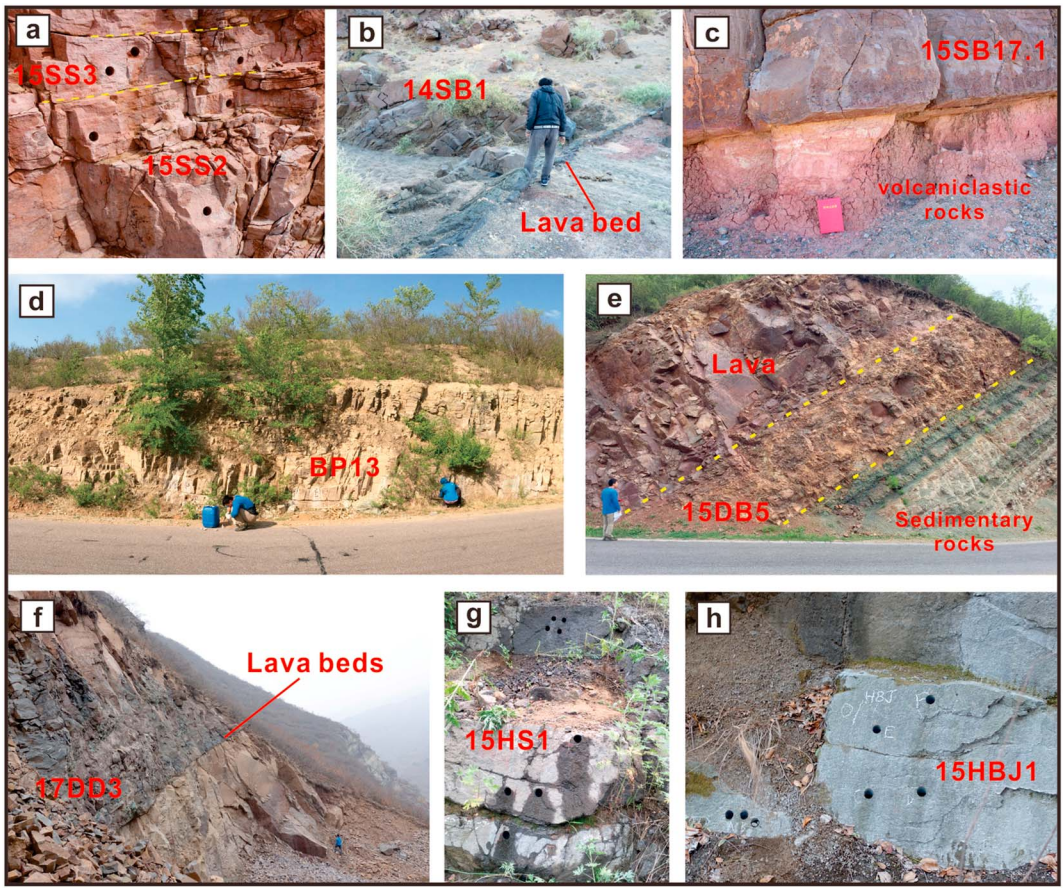


Figure 5. Photographs of sampling outcrops. (a) sandstones of Sharilyn Fm (~155 Ma) in southern Mongolia; (b) basalts of section I of Tsagantsav Fm (~130 Ma) in southern Mongolia; (c) basalts with red volcaniclastic rocks interbed of section II of Tsagantsav Fm (~130 Ma) in southern Mongolia; (d) sandstones of upper Tuchengzi Fm (~140 Ma) in Beipiao, NCB; (e) lavas and sedimentary rocks of section I of upper Dadianzi Fm (~130 Ma) in Luanping, NCB; (f) lavas and volcaniclastic rocks of section II of upper Dadianzi Fm (~130 Ma) in Luanping, NCB; (g) vesicular basalts of the lower Yixian Fm in Sihetun, Beipiao, NCB; (h) vesicular basalts of the lower Yixian Fm in Huangbanjigou, Beipiao, NCB. NCB = North China Block.

geochronological sampling sites, we have carried out paleomagnetic sampling for the top of Dadianzi Fm volcanic and sedimentary rocks in two sections (a total of 119 core samples, including section I: 88 samples from 15 sites, and section II: 30 samples from 5 sites; Table 1 and Figures 3c and 4, Column D).

In the Beipiao Basin, volcanic rocks of the Yixian Fm are widely distributed in Sihetun village and the surrounding area (Figure 3). It contains three volcano-sedimentary cycles that rest unconformably over the Tuchengzi Fm (Figure 2). The lacustrine beds of the first volcano-sedimentary cycle in the lower part of the Yixian Fm have yielded a wide range of well-preserved fossils (Ji et al., 2004; Swisher et al., 1999; S. Wang et al., 2001; Y. Q. Wang et al., 2016), which can be correlated with the upper fossil-bearing layers of the Dadianzi Fm (Figure 2; Liu et al., 2002; Pang et al., 2006; Tian et al., 2004; H. Zhang, Guo, et al., 2008). In the Sihetun and Huangbanjigou areas, we collected 122 paleomagnetic samples from 15 sites (Sihetun: 82 samples from nine sites; Huangbanjigou: 40 samples from six sites; Figures 3d and 4, Column E) from the vesicular basalts of the Xiatulaigou Member at the bottom of the Yixian Fm, which has been dated at 130.5 ± 0.5 and 129.7 ± 0.5 Ma using the $^{40}Ar/^{39}Ar$ method by Peng et al. (2003) and Chang et al. (2009), respectively.

3. Rock-Magnetic Results

To identify the magnetic minerals and ensure the reliability of the paleomagnetic data, isothermal remanent magnetization (IRM), back-field demagnetization of saturation IRM (SIRM), and thermal demagnetization of the three-axis IRMs (Lowrie, 1990) were measured for representative specimens at Paleomagnetism and Environmental Magnetism Laboratory of the China University of Geosciences, Beijing. Fields of 2.5, 0.4 and

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Table 1
Site-Mean Values and Statistical for the High-Temperature Component of Late Jurassic and Early Cretaceous From Amuria and North China Blocks

Site	n/N	Strike/dip (°)	In geographic coordinates				In stratigraphic coordinates						
			D (°)	I (°)	k (°)	α_{95} (°)	D (°)	I (°)	k (°)	α_{95} (°)	Plat (°N)	Plon (°E)	dp/dm
Amuria Block (AMU)													
Sharilyn Formation (~155 Ma) sandstone													
Section I (44.518°N, 109.266°E)													
14SS1	7/8	265/13	21.9	63.2	109.5	5.8	14.0	51.2	109.5	5.8	73.3	243.7	7.2/9.1
14SS2	7/13	271/22	32.3	58.6	67.0	7.4	21.3	38.7	67.0	7.4	61.3	244.6	5.2/8.8
14SS3	9/11	270/6	30.6	61.4	41.2	8.1	25.9	56.1	41.2	8.1	68.9	212.2	8.4/11.7
14SS4.1	5/5	245/16	50.3	62.7	30.3	14.1	26.1	55.3	30.3	14.1	68.4	213.9	14.3/18.3
14SS4.2	4/5	245/16	35.0	62.7	62.6	11.7	15.5	52.3	62.6	11.7	73.3	238.0	11.0/16.1
Section II (44.509°N, 109.281°E)													
15SS1.1	8/8	235/8	15.4	64.0	90.7	5.8	5.1	58.3	90.7	5.8	83.3	253.0	6.3/8.6
15SS1.2	6/6	235/8	13.4	62.6	394.0	3.4	4.0	56.8	394.0	3.4	82.3	265.0	3.6/4.9
15SS2	8/8	235/8	19.4	60.6	57.6	7.4	9.6	55.4	57.6	7.4	78.7	245.6	7.5/10.6
15SS3	6/10	235/8	200.4	−56.3	23.4	14.1	191.9	−51.2	23.4	14.1	−74.3	68.9	12.9/19.1
15SS4.1	7/7	240/11	26.4	61.5	82.9	6.7	12.9	54.3	82.9	6.7	76.2	239.2	6.6/9.4
15SS4.2	6/6	240/11	25.4	69.6	198.8	4.8	7.6	61.9	198.8	4.8	84.3	210.8	5.8/7.4
15SS5	8/8	240/11	30.8	58.2	36.8	9.3	18.0	51.8	36.8	9.3	71.5	233.9	8.7/12.7
15SS6	11/11	240/11	17.7	65.9	67.5	5.6	4.2	57.6	67.5	5.6	83.0	261.2	6.0/8.2
15SS7	7/7	237/16	53.6	58.0	141.0	5.1	30.5	53.7	141.0	5.1	64.5	212.4	5.0/7.1
15SS8.1	5/5	237/16	46.4	61.9	51.1	10.8	21.8	55.5	51.1	10.8	71.4	218.9	11.0/15.4
15SS8.2	6/6	237/16	40.3	60.4	64.1	8.4	18.5	52.9	64.1	8.4	71.9	230.6	8.0/11.6
Mean I + II[a]	110/124	*N* = 16	30.3	62.3	146.0	3.1	15.8	54.2	139.5	3.1	74.7	232.5	A_{95} = 3.7°
Tsagantsav Formation (~130 Ma) basalts													
Section I (44.465°N, 109.381°E)													
14SB1	10/10	111/26	199.7	−28.1	121.7	4.4	199.0	−54.1	121.7	4.4	−72.4	46.9	4.3/6.2
14SB2.1	8/8	111/26	189.7	−32.5	10.7	5.4	183.0	−57.7	10.7	5.4	−83.5	88.2	5.8/7.9
14SB2.2	8/8	111/26	200.9	−35.1	147.9	4.6	200.8	−61.1	147.9	4.6	−74.7	20.7	5.4/7.1
14SB3.1	9/9	111/26	194.8	−34.4	204.9	3.6	190.6	−60.2	204.9	3.6	−81.5	39.0	4.1/5.5
14SB3.2	10/10	111/26	202.7	−31.7	96.7	4.9	203.7	−57.7	96.7	4.9	−71.3	30.2	5.3/7.2
14SB4.1	6/6	111/26	200.3	−34.8	1383.0	1.8	199.8	−60.8	1383.0	1.8	−75.3	22.9	2.1/2.7
14SB4.2	9/9	111/26	199.6	−35.5	718.9	1.9	198.5	−61.4	718.9	1.9	−76.5	21.1	2.3/2.9
14SB5	11/11	111/26	201.5	−39.3	31.8	8.2	201.9	−65.3	31.8	8.2	−74.5	0.7	10.8/13.3
14SB6.1	7/7	111/26	196.5	−33.6	110.8	5.8	193.7	−59.5	110.8	5.8	−79.1	36.9	6.5/8.7
14SB6.2	7/7	111/26	194.5	−37.5	118.2	5.6	189.5	−63.2	118.2	5.6	−83.2	14.0	7.0/.8.0
14SB7.1	5/5	111/26	193.6	−36.9	221.9	5.1	188.0	−62.6	221.9	5.1	−84.2	21.5	6.2/8.0
14SB7.2	10/10	111/26	197.1	−38.7	265.1	3.0	193.9	−64.6	265.1	3.0	−80.1	2.8	3.9/4.8
14SB8	11/11	80/15	8.6	45.2	685.9	1.7	15.9	59.2	685.9	1.7	77.4	214.7	1.9/2.5
14SB9	6/6	80/15	17.9	49.0	311.2	3.8	30.1	61.5	311.2	3.8	68.2	193.6	4.5/5.9
14SB10	6/6	80/15	356.3	49.8	162.6	4.0	359.5	64.6	162.6	4.0	88.0	99.7	5.2/6.4
15SB18	11/11	111/26	204.6	−32.3	40.5	7.3	206.8	−58.2	40.5	7.3	−69.3	25.7	8.0/10.8
15SB19	8/8	111/26	198.3	−41.5	26.5	11.0	195.7	−67.4	26.5	11.0	−77.9	345.3	15.2/18.3
15SB20	7/7	111/26	212.7	−32.9	24.2	4.5	219.7	−58.0	24.2	4.5	−60.0	16.5	4.9/6.6
15SB21	5/5	111/26	211.2	−35.6	157.1	6.1	218.2	−60.9	157.1	6.1	−62.3	11.0	7.1/9.3
15SB22	5/5	80/15	13.3	48.8	374.9	4.0	23.9	62.1	374.9	4.0	72.8	194.6	4.8/6.2
Section II (44.471°N, 109.414°E)													
15SB11	8/8	30/8	4.9	65.5	361.7	2.9	22.8	67.8	361.7	2.9	73.5	168.9	4.1/4.9
15SB12	7/7	30/8	10.2	64.2	269.4	3.4	27.1	65.8	269.4	3.4	71.0	178.9	4.5/5.5
15SB13	10/10	30/8	9.3	63.4	622.5	1.9	25.6	65.2	622.5	1.9	72.0	181.4	2.5/3.1
15SB14	9/9	30/8	7.2	64.0	38.8	8.4	23.9	66.0	38.8	8.4	73.1	177.8	11.2/13.7
15SB15	6/6	30/8	11.3	62.6	338.3	3.6	27.0	64.1	338.3	3.6	71.0	185.7	4.6/5.7
15SB16	8/8	30/8	12.3	61.9	248.5	3.5	27.6	63.3	248.5	3.5	70.5	188.6	4.4/5.5
15SB17.1	5/5	73/10	206.2	−55.2	590.3	3.2	218.6	−61.8	590.3	3.2	−62.3	8.7	3.8/5.0
15SB17.2	6/6	73/10	205.3	−51.6	133.7	5.8	215.9	−58.4	133.7	5.8	−62.9	18.2	6.4/8.6
Mean I + II[b]	218/218	*N* = 28	17.3	44.5	36.7	4.6	21.6	62.2	190.3	2.0	74.6	194.7	A_{95} = 2.9°
North China Block (NCB)													
Tuchengzi Formation (~140 Ma) sandstones													
BP1–7 (41.745°N, 120.803°E), BP8–17 (41.604°N, 120.808°E), BP18 (41.682°N, 120.748°E), BP19–21 (41.551°N, 120.831°E)													
BP1	4/5	257/13	39.8	82.0	110.0	8.8	18.9	62.2	110.0	8.8	76.0	197.3	10.7/13.7
BP2	6/6	257/13	35.6	64.8	243.8	4.3	20.0	54.9	243.8	4.3	73.2	226.3	4.3/6.1
BP3	7/10	240/11	40.8	64.6	130.7	5.3	22.5	59.3	130.7	5.3	73.0	208.8	5.9/7.9

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Table 1 *(continued)*

Site	n/N	Strike/dip (°)	In geographic coordinates				In stratigraphic coordinates						
			D (°)	I (°)	k (°)	α_{95} (°)	D (°)	I (°)	k (°)	α_{95} (°)	Plat (°N)	Plon (°E)	dp/dm
BP4	7/11	257/13	27.1	64.2	48.0	8.8	15.1	53.4	48.0	8.8	75.8	239.1	8.5/12.2
BP5	6/6	240/11	41.2	63.5	573.6	2.8	23.5	58.4	573.6	2.8	72.0	211.4	3.1/4.1
BP6	5/5	240/11	45.3	62.0	507.5	3.4	27.9	57.5	507.5	3.4	68.4	211.2	3.6/5.0
BP7	5/7	240/11	31.8	55.8	143.9	6.4	19.9	49.6	143.9	6.4	70.4	239.7	5.7/8.5
BP8	4/6	89/11	175.8	−50.8	36.1	15.5	174.7	−61.8	36.1	15.5	−85.8	232.2	18.6/24.0
BP9	4/5	89/11	177.2	−52.5	79.0	10.4	176.5	−63.5	79.0	10.4	−85.7	265.8	13.0/16.5
BP10	4/6	97/10	184.7	−53.3	62.6	11.7	183.9	−63.3	62.6	11.7	−85.7	340.9	14.6/18.5
BP11	5/5	97/10	185.5	−52.5	366.9	4.0	185.0	−62.5	366.9	4.0	−85.7	357.7	4.9/6.2
BP12	5/5	104/11	3.2	58.3	48.5	11.1	358.0	69.0	48.5	11.1	79.0	114.4	16.0/18.9
BP13	4/7	89/11	178.1	−47.5	35.2	15.7	177.8	−58.5	35.2	15.7	−87.1	156.6	17.3/23.3
BP14	4/6	89/11	176.0	−48.7	80.5	10.3	175.0	−59.6	80.5	10.3	−86.1	195.3	11.7/15.5
BP15	4/5	97/10	196.0	−47.4	76.1	10.6	198.2	−57.3	76.1	10.6	−75.6	39.6	11.3/15.5
BP16	5/6	97/10	184.1	−47.9	163.6	6.0	182.9	−58.1	163.6	6.0	−86.4	81.8	6.5/8.9
BP17	5/7	104/11	16.4	60.5	60.7	9.9	17.8	71.4	60.7	9.9	71.5	153.4	15.2/17.3
BP18	6/8	98/20	188.8	−37.7	129.9	5.9	189.2	−57.7	129.9	5.9	−82.2	53.1	6.4/8.7
BP19	7/7	245/10	359.2	74.0	71.2	7.2	350.2	64.6	71.2	7.2	81.4	69.1	9.3/11.6
BP20	4/5	245/10	3.1	67.3	96.5	9.4	355.1	58.1	96.5	9.4	85.3	355.9	10.2/13.9
BP21	6/8	228/10	217.8	−55.4	58.9	8.8	204.6	−52.5	58.9	8.8	−68.8	46.3	8.3/12.1
Mean[c]	107/136	$N = 21$	14.5	58.3	39.2	5.1	10.1	60.2	103.4	3.1	82.7	208.6	$A_{95} = 4.3°$
Dadianzi Formation (~130 Ma) volcanic rocks + sedimentary rocks													
Section I: 15DB1–5 (volcanic rocks, 40.870°N, 117.282°E), 15DB6–7 and 17DB1–8 (sedimentary rocks, 40.869°N, 117.283°E)													
15DB1	4/6	240/37	277.8	−59.0	123.5	8.3	207.1	−61.0	123.5	8.3	−69.7	14.9	9.7/12.7
15DB2.1	5/5	240/37	298.1	−67.2	158.3	6.1	185.0	−69.1	158.3	6.1	−77.8	311.7	8.8/10.4
15DB2.2	6/6	240/37	282.5	−62.6	187.9	4.9	199.9	−63.5	187.9	4.9	−74.9	4.5	6.1/7.8
15DB3	5/5	240/37	287.2	−64.7	71.7	9.1	194.7	−65.6	71.7	9.1	−77.4	349.0	12.0/14.8
15DB4	4/7	240/37	270.9	−70.6	48.0	13.4	183.5	−58.9	48.0	13.4	−87.1	50.6	14.9/20.0
15DB5	6/6	233/43	277.7	−54.8	27.1	13.1	200.5	−60.9	27.1	13.1	−74.6	376.5	15.3/20.0
15DB6	4/6	235/29	73.4	71.4	100.7	9.2	5.2	57.5	100.7	9.2	85.1	240.0	9.9/13.5
15DB7	7/7	235/29	75.6	68.7	39.7	9.7	10.0	61.2	39.7	9.7	82.4	193.3	11.4/14.9
17DB1	5/5	235/26	63.4	71.7	747.8	2.8	4.6	60.9	747.8	2.8	86.4	188.6	3.3/4.3
17DB2	4/6	235/26	69.0	71.3	159.4	7.3	7.0	62.3	159.4	7.3	84.1	177.2	8.9/11.4
17DB3	6/6	235/26	72.2	62.9	43.3	10.3	23.3	59.2	43.3	10.3	72.3	202.5	11.5/15.4
17DB5	8/8	235/26	76.0	69.4	46.6	8.2	12.9	63.7	46.6	8.2	79.6	177.6	10.3/13.0
17DB6	5/5	245/24	45.0	72.8	651.5	3.0	5.0	56.3	651.5	3.0	84.4	251.5	3.1/4.3
17DB7	5/5	235/23	55.8	66.2	60.7	9.9	13.9	57.6	60.7	9.9	79.0	216.6	10.6/14.5
17DB8	6/6	250/23	52.5	75.1	167.0	5.2	8.7	59.2	167.0	5.2	83.3	212.0	5.8/7.8
Section II: 17DD3 (basalts, 40.928°N, 117.381°E), 17DD4–5 (sandstones, 40.927°N, 117.383°E), 17DD6–7 (andesites, 40.928°N, 117.384°E)													
17DD3	3/6	224/23	240.7	−70.6	999.9	3.2	182.6	−64.9	999.9	3.2	−83.8	314.0	4.2/5.2
17DD4	7/8	234/16	26.9	64.5	217.7	4.1	5.2	54.4	217.7	4.1	82.7	261.4	4.1/5.8
17DD5	5/5	234/16	44.6	65.7	53.1	10.6	16.1	59.0	53.1	10.6	77.7	207.5	11.8/15.8
17DD6	5/5	210/25	61.9	58.9	85.5	8.3	15.4	63.0	85.5	8.3	78.2	184.9	10.3/13.0
17DD7	5/6	210/25	64.4	56.9	345.4	4.1	20.6	62.8	345.4	4.1	74.5	188.3	5.0/6.4
Mean I + II[d]	105/119	$N = 20$	74.2	68.0	55.4	4.4	12.0	61.2	263.4	2.0	80.9	193.1	$A_{95} = 2.8°$
Yixian Formation (~130 Ma) basalts													
Section III (Sihetun) - (41.592°N, 120.802°E)													
15HS1	8/8	35/18	176.8	−54.8	144.6	4.6	204.2	−62.5	144.6	4.6	−72.1	375.4	5.6/7.2
15HS2	12/12	35/18	167.5	−48.1	185.2	3.2	187.2	−59.3	185.2	3.2	−84.4	43.7	3.6/4.8
15HS3	12/12	35/18	162.6	−47.6	281.5	2.6	180.7	−60.2	281.5	2.6	−89.3	72.3	3.0/3.9
15HS4	14/14	35/18	165.4	−46.6	267.1	2.4	183.5	−58.5	267.1	2.4	−86.4	71.4	2.6/3.6
15HS5	7/7	35/18	166.4	−51.3	325.5	3.4	188.6	−62.6	325.5	3.4	−83.3	367.3	4.2/5.3
17HS1	6/6	35/18	158.4	−53.8	124.9	6.0	180.9	−66.9	124.9	6.0	−82.0	305.0	8.2/9.9
17HS2	8/8	35/18	165.9	−50.4	789.8	2.0	187.2	−61.9	789.8	2.0	−84.5	372.4	2.4/3.1
17HS5	8/8	35/18	186.4	−50.0	205.5	3.9	209.3	−55.5	205.5	3.9	−66.6	34.9	4.0/5.6
17HS6	7/7	35/18	176.7	−45.8	111.1	5.8	195.9	−54.6	111.1	5.8	−76.0	53.1	5.8/8.2
Mean	82/82	$N = 9$	169.6	−50.1	162.4	4.1	191.3	60.6	162.4	4.1	−81.7	26.0	$A_{95} = 5.7°$
Section IV (Huangbanjigou) - (41.608°N, 120.830°E)													
17HBJ1	7/7	61/11	7.4	54.7	84.0	6.6	19.7	62.9	84.0	6.6	75.4	193.5	8.1/10.4
17HBJ2	7/7	61/11	356.4	50.9	52.9	8.4	4.3	60.5	52.9	8.4	86.8	211.9	9.7/12.8
17HBJ3	3/5	61/11	3.4	54.4	124.0	11.1	14.6	63.1	124.0	11.1	78.9	190.3	13.8/17.5

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Table 1 *(continued)*

Site	n/N	Strike/dip (°)	In geographic coordinates				In stratigraphic coordinates						
			D (°)	I (°)	k (°)	α_{95} (°)	D (°)	I (°)	k (°)	α_{95} (°)	Plat (°N)	Plon (°E)	dp/dm
17HBJ5	5/7	61/11	15.3	55.2	118.6	7.1	29.5	62.1	118.6	7.1	68.3	196.2	8.6/11.0
17HBJ6	5/7	61/11	14.4	54.8	143.5	6.4	28.2	61.9	143.5	6.4	69.2	197.2	7.7/9.9
17HBJ8	7/7	50/19	353.8	43.2	349.8	3.2	9.2	57.6	349.8	3.2	82.2	233.4	3.4/4.7
Mean	34/40	$N = 6$	4.5	52.5	124.4	6.0	17.3	61.7	234.6	4.4	77.1	199.3	$A_{95} = 6.5°$
Mean III + IV[e]	116/122	$N = 15$	355.3	51.3	100.1	3.8	13.6	61.0	184.7	2.8	79.9	202.6	$A_{95} = 4.0°$
Mean I + II + III + IV	221/241	$N = 35$	29.3	66.5	13.6	6.8	12.7	61.2	228.5	1.6	80.5	197.4	$A_{95} = 2.3°$

Note. n/N = number of samples used to calculate mean/total samples demagnetized; Strike/Dip = right hand strike/dip of the strata; D/I = declination/inclination; k = the precision parameter; α_{95} = the radius that the mean direction lies within 95% confidence; Plat/Plon = latitude/longitude of paleopoles; dp/dm = semiaxes of elliptical error of the pole at a probability of 95%.
[a](1) Fold test is inconclusive at the 95% and 99% confidence (McElhinny, 1964): Ks/Kg = 1.05 < F(2*(n2 − 1),(n1 − 1)) at 5% and 1% point = 1.84 and 2.38, respectively; (2) fold test is positive at the 95% confidence (McFadden,1990), critical Xi at 95% = 4.66, Xi2 IS = 5.13, Xi2 TC = 2.43; (3) McFadden and McElhinny (1990) reversal test, angle between the two averages $\gamma = 4.1° < \gamma_{critical} = 13.3°$ indicates a C class result. [b](1) Fold test is positive at the 95% and 99% confidence (McElhinny, 1964): Ks/Kg = 5.18 > F(2*(n2 − 1),(n1 − 1)) at 5% and 1% point = 1.57 and 1.90, respectively; (2) fold test is also positive at the 95% and 99% confidence (McFadden,1990), critical Xi at 95% = 6.16, at 99% = 8.70; Xi1 IS = 10.76, Xi1 TC = 0.79; Xi2 IS = 25.14, Xi2 TC = 4.71; (3) McFadden and McElhinny (1990) reversal test, angle between the two averages $\gamma = 3.1° < \gamma_{critical} = 4.1°$ indicates an A class result. [c](1) Fold test is positive at the 95% and 99% confidence (McElhinny, 1964): Ks/Kg = 2.64 > F(2*(n2 − 1),(n1 − 1)) at 5% and 1% point = 1.69 and 2.11, respectively; (2) fold test is also positive at the 95% and 99% confidence (McFadden,1990), critical Xi at 95% = 5.34, at 99% = 7.48; Xi1 IS = 14.24, Xi1 TC = 0.94; Xi2 IS = 14.86, Xi2 TC = 0.91; (3) McFadden and McElhinny (1990) reversal test, angle between the two averages $\gamma = 4.6° < \gamma_{critical} = 6.2°$ indicates a B class result. [d](1) Fold test is positive at the 95% and 99% confidence (McElhinny, 1964): Ks/Kg = 4.76 > F(2*(n2 − 1),(n1 − 1)) at 5% and 1% point = 1.72 and 2.16, respectively; (2) fold test is also positive at the 95% and 99% confidence (McFadden,1990), critical Xi at 95% = 5.21, at 99% = 7.30; Xi2 IS = 14.48, Xi2 TC = 2.73; (3) McFadden and McElhinny (1990) reversal test, angle between the two averages $\gamma = \gamma_{critical} = 4.0°$ indicates an A class result. [e](1) Fold test is inconclusive at the 95% and 99% confidence (McElhinny, 1964): Ks/Kg = 1.84 < F(2*(n2 − 1), (n1 − 1)) at 5% and 1% point = 1.88 and 2.47, respectively; (2) fold test is positive at the 99% confidence (McFadden,1990), critical Xi at 99% = 6.31; Xi2 IS = 9.83, Xi2 TC = 5.69; (3) McFadden and McElhinny (1990) reversal test, angle between the two averages $\gamma = 3.1° < \gamma_{critical} = 5.8°$ indicates a B class result.

0.12 T were applied in turn along the *z*, *y*, and *x* axes of the specimens using the IM10–30 pulse magnetizer, respectively. Then, stepwise thermal demagnetization up to 680 °C was conducted. The IRM was measured using an AGICO JR-6A spinner magnetometer.

For the sandstone specimens of the Sharilyn Fm and Tuchengzi Fm, the IRM acquisition curves show a rapid increase below 150 mT and acquire 80% of the SIRM (Figures 6a and 6c), which suggests that low-coercivity magnetic carriers are dominant. The IRM intensity increases slower above 150 mT, but does not reach saturation value until at 2.5 T (Figures 6a and 6c), revealing the presence of high-coercivity magnetic carriers. Progressive back-demagnetization of SIRM shows that the remanent coercive force is less than 50 mT (Figures 6a and 6c), indicating that the presence of low coercivity minerals in these samples is highlighted. Stepwise thermal demagnetization of three orthogonal IRM components (Lowrie, 1990) shows that the soft component (0.12 T) was unblocked around 580 °C and that the hard component (2.5 T) was unblocked at 675 °C (Figures 6b and 6d). Thus, these results reveal the presence of both magnetite and hematite, but magnetite is the predominant magnetic carriers in the sandstones.

For the lava specimens from the Tsagantsav Fm and Dadianzi/Yixian Fm, the IRM acquisition curve shows a rapid increase below 100 mT, and saturation is fully reached at 200 mT, which, combined with the finding that the maximum coercive force is less than 20 mT (Figures 6e and 6g), reveals that low-coercivity magnetic carriers are dominant. Stepwise thermal demagnetization of three orthogonal IRM components (Lowrie, 1990) shows that each component, hard (2.5 T), medium (0.4 T), and soft (0.12 T), has a similar unblocking temperature around 580 °C (Figures 6f and 6h). These rock-magnetic results reveal that the remanent magnetizations of the lava specimens are mainly carried by magnetite.

4. Paleomagnetic Results

Based on the rock magnetic analyses, all specimens were subjected to stepwise thermal demagnetization. Most volcanic specimens were heated up to 580 °C with a few up to 620 °C and all the sedimentary specimens were heated up to 680 °C. The temperature intervals generally ranged from a maximum of 100 °C for lower temperature steps to a minimum of 5 °C for higher temperature steps. The remanent magnetizations were measured using a 2G-755-4K cryogenic magnetometer for the sedimentary rocks and an AGICO

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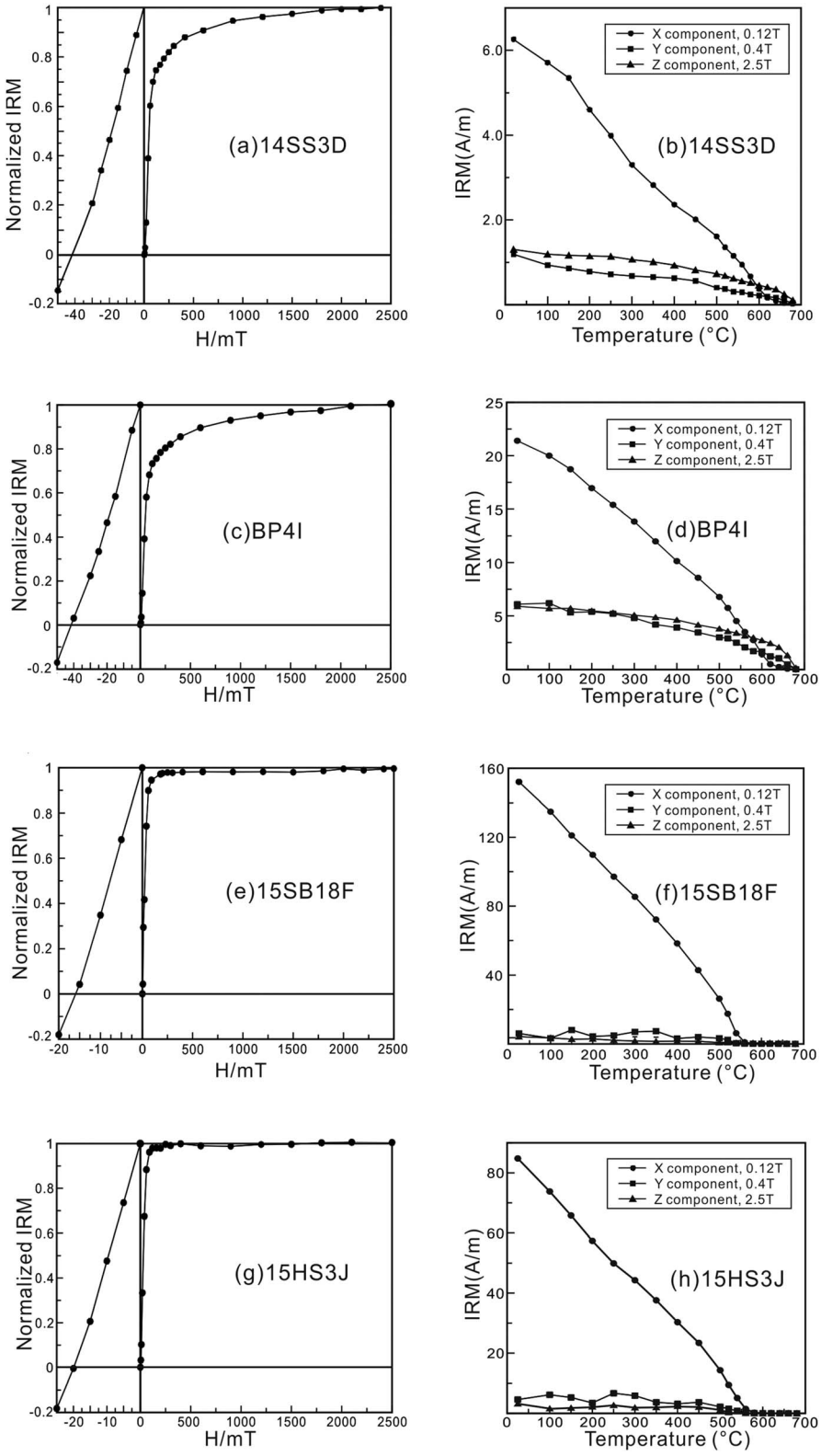


Figure 6. IRM and back-field demagnetization of saturation IRM acquisition curves of representative samples (a, c, e, and g); thermal demagnetization of the three-axis IRM showing unblocking temperatures around 580 or 680 °C (b, d, f and h). The representative samples for 14SS3D and BP4I from the sandstones of Sharilyn Fm (~155 Ma) and Tuchengzi Fm (~140 Ma), respectively; and 15SB18F and 15HS3J from the basaltic lavas of Tsagantsav Fm (~130 Ma) and Yixian Fm, respectively. IRM = Isothermal Remanent Magnetization.

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JR-6A spinner magnetometer for the strongly magnetized volcanic rocks. The measurements were performed within a micrometal shielded room at Paleomagnetism and Environmental Magnetism Laboratory of China University of Geosciences, Beijing with residual fields of less than 200 nT. Remanent magnetization directions were analyzed using principal component analysis (Kirschvink, 1980) for all specimens. Site-mean and overall-mean directions were calculated using Fisher (1953) statistics. Paleomagnetic data processing and interpretations were done using the computer software packages of Cogné (2003) and Enkin (1990).

4.1. Southern Mongolia, Amuria

4.1.1. The Upper Jurassic (~155 Ma) Sharilyn Fm

Of the 124 specimens that underwent stepwise thermal demagnetization, 110 specimens revealed useful magnetic signals (Table 1), whereas the remaining specimens were not accepted because of coarse granularity or breaking up during heating. The natural remanent magnetization (NRM) intensities ranged from 2 to 80 mA/m. Representative Zijderveld plots (after Zijderveld, 1967) are shown in Figures 7a–7c. A low-temperature component (LTC) could be identified for most specimens between room temperature and 250 °C. The LTCs in situ cluster near the local modern geomagnetic field direction (D = −5.0°, *I* = 64.6°, IGRF online data; Figure 8a), revealing that the LTC is a viscous remanent magnetization (VRM) of the present geomagnetic field. After removal of the LTC, a stable high-temperature component (HTC) decayed toward the origin near the unblocking temperatures of 580 or 640 °C. The HTC was dual-polarity, with magnetization directed northeastward and moderately down as normal polarity and southwestward and moderately up as reversal polarity (Figure 9a). The reversal polarity direction was only at site-15SS3, where 40-cm-thick strata were sampled (Figure 4, Column A). The site-mean direction for 16 sites is Dg = 30.3°, Ig = 62.3°, k = 146.0, and α_{95} = 3.1° in situ, which is significantly different from the modern field direction, and Ds = 15.8°, Is = 54.2°, k = 139.5, and α_{95} = 3.1° after tilt correction (Figure 9a and Table 1).

All of the site-level VGPs of the Sharilyn Fm sandstones show Fisher distribution; the results of the "quantile-quantile" plot test (Fisher et al., 1987) are positive (Mu = 1.152 < 1.207, Me = 0.686 < 1.094). Furthermore, we used the widely accepted elongation/inclination (E/I) method (Tauxe & Kent, 2004) to further test the possible inclination shallowing for the sedimentary rocks. We calculated the flattening correction factor "f value" at sample level (*N* = 110) by using the software of Lisa Tauxe's PmagPy Cookbook (https://earthref.org/PmagPy/cookbook/). The f value was 0.9 and inclination was corrected from 54.4° to 55.7°, with the 95% confidence interval between 54.1° and 68.6°. These observations suggest that the results from the sedimentary rocks were not significantly affected by inclination shallowing. We thus use the original HTCs to calculate the pole.

The dual polarity directions of all the site-level HTCs can pass a positive class C reversal test at the 95% probability level: $\gamma_o = 4.1 < \gamma_{critical} = 13.3$ (McFadden & McElhinny, 1990). Furthermore, they can also pass the positive fold test of McFadden (1990) at the 95% confidence level (Table 1). These findings all indicate that the HTC is a prefolding primary magnetization. Therefore, we obtained the mean pole at 74.7°N/232.5°E (A_{95} = 3.7°) by averaging all of the site-level VGPs for the ~155 Ma Sharilyn Fm.

4.1.2. The Lower Cretaceous (~130 Ma) Tsagantsav Fm

All 218 specimens from 28 sites, which were obtained from the basalt flows of two sections of the Tsagantsav Fm (~130 Ma), underwent stepwise thermal demagnetization and revealed useful magnetic signals. NRM intensities ranged from 0.2 to 6 A/m. The representative Zijderveld plots (after Zijderveld, 1967) are shown in Figures 7d–7h. In most cases, an LTC could be defined below 300 °C. The LTCs in situ are clustered around the local modern geomagnetic field direction (D = −5.2°, *I* = 64.6°, IGRF online data) and are considered a VRM of the present magnetic field (Figure 8b). After removal of the LTC, a stable HTC decayed toward the origin near the unblocking temperatures of 580 °C. Each section has dual polarity (Figure 9b). The in situ site-mean direction of the HTCs for 28 sites from two sections is Dg = 17.3°, Ig = 44.5°, k = 36.7, α_{95} = 4.6°, which is obviously different from that of the local geomagnetic field direction. After tilt correction, the site-mean direction is Ds = 21.6°, Is = 62.2°, k = 190.3, α_{95} = 2.0° (Figure 9b and Table 1). These site-level HTCs can pass the fold tests of both McElhinny (1964) and McFadden (1990) at the 95% and 99% confidence levels (Table 1), and they can also pass a reversal test (McFadden & McElhinny, 1990) at the 95% confidence level (class A) with an angle between the two averages $\gamma_o = 3.1 < \gamma_{critical} = 4.1$. Thus, the HTC of the Tsagantsav Fm basalts represents a prefolding primary magnetization.

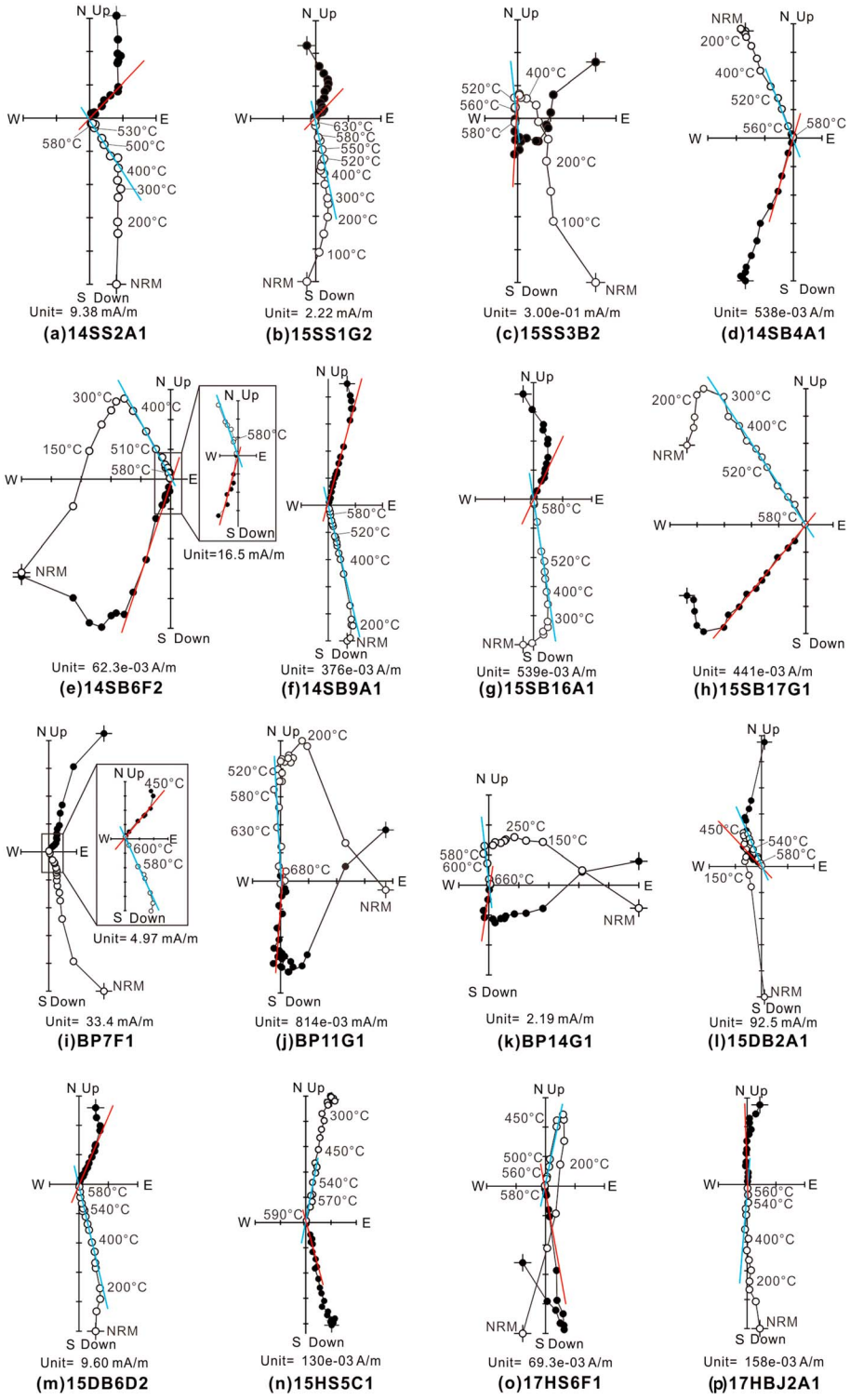


Figure 7. Typical demagnetization characteristics of samples, in geographic coordinates. Solid/open symbols of the orthogonal plots represent the projections onto the horizontal/vertical plane. (a)–(c), Sharilyn Fm (~155 Ma) sandstones; (d)–(h), Tsagantsav Fm (~130 Ma) basalts; (i)–(k), Tuchengzi Fm (~140 Ma) sandstones; (i) and (m) from the Dadianzi Fm (~130 Ma) basalt and sandstone, respectively; (n)–(p), Yixian Fm (~130 Ma) basalts. NRM = natural remanent magnetization.

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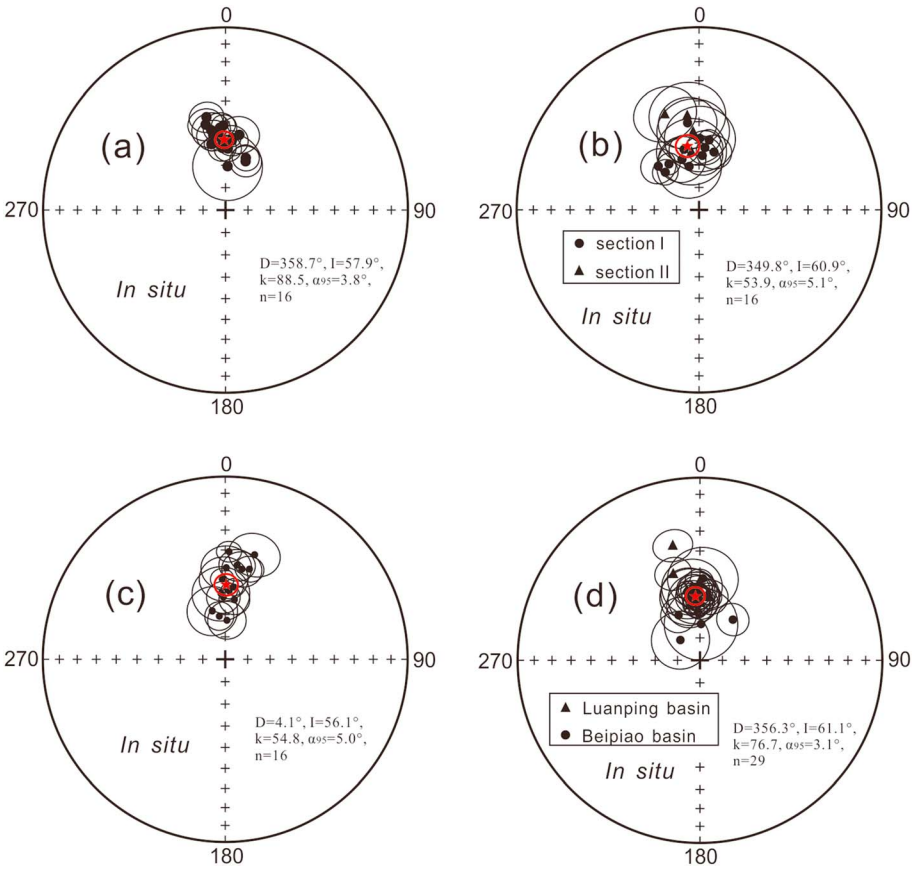


Figure 8. Equal-area stereographic projections of the site-mean directions of the low temperature component. (a) Sharilyn Fm (~155 Ma), (b) Tsagantsav Fm (~130 Ma), (c) Tuchengzi Fm (~140 Ma), and (d) Dadianzi and Yixian Fms (~130 Ma).

Because the paleomagnetic records of each volcanic flow provide only spot readings of geomagnetic field behavior (Tauxe, 1993), and paleosecular variation of the geomagnetic field can at any time and location easily result in deviation of tens of degrees in the geomagnetic field direction (Deenen et al., 2011), it is necessary to test whether the ~130 Ma volcanic rocks of the Tsagantsav Fm have averaged out the PSV. There are four types of evidence: (1) the sampling of lava flows spans a long interval of time because many lava flows are interbedded with sedimentary rocks in two sections (Figure 4; Column B); (2) HTC directions include antipodal dual polarities and pass the reversal test; (3) the results of the quantile-quantile plot test (Fisher et al., 1987) are positive (Mu = 0.893 $<$ 1.207, Me = 0.567 $<$ 1.094) for all site-mean VGPs of the Tsagantsav Fm, indicating that the VGPs are Fisher-distributed; and (4) the value of A_{95}, which was obtained from the VGPs of 218 lava specimens, is 1.5, which is consistent with an N-dependent A_{95} envelope with a 95% confidence interval (1.39, 2.76) proposed by Deenen et al. (2011, 2014). Recently, this new model has been widely accepted (e.g., Y. Ma et al., 2016; Palencia-Ortas et al., 2011; Ren et al., 2016). Above all, the paleomagnetic results from the lavas (28 independent sites) of the Tsagantsav Fm are free of PSV influence.

Based on the field tests and PSV analyses, we obtained the mean pole at 74.6°N/194.7°E (A_{95} = 2.9°) by averaging all of the site-level VGPs for the ~130 Ma Tsagantsav Fm (Table 1).

4.2. Yanshan Belt, North China

4.2.1. The Upper Jurassic–Lower Cretaceous (~140 Ma) Tuchengzi Fm

All the 136 specimens from Tuchengzi Fm sandstones underwent stepwise thermal demagnetization, and their NRM intensities ranged from 1.7 to 220 mA/m. The representative Zijderveld plots (after Zijderveld, 1967) are shown in Figures 7i–7k. For most specimens, a LTC could be identified between room temperature and 300 °C. The in situ LTCs cluster near the local modern geomagnetic field direction (D = −8.3°, *I* = 60.1°, IGRF online data; Figure 8c), which suggests that the LTC is a VRM of the present magnetic field. After

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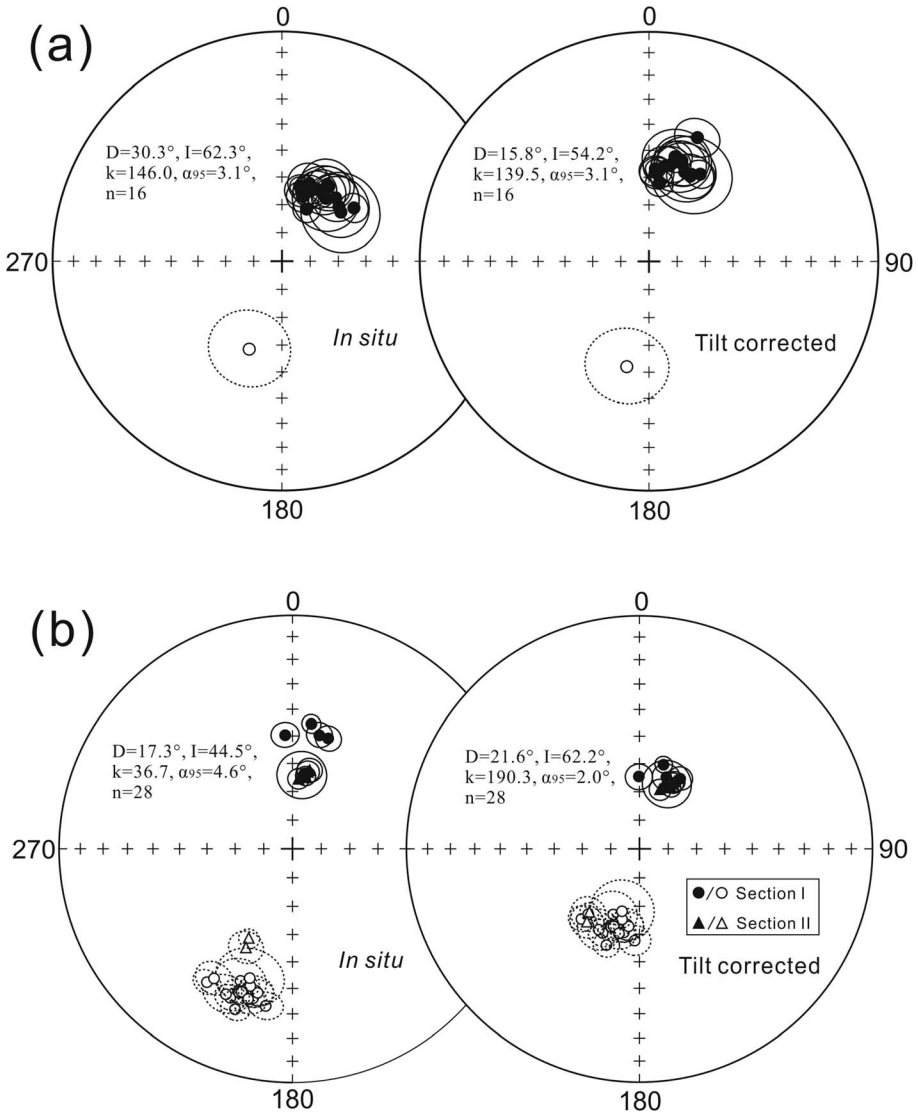


Figure 9. Equal-area stereographic projections of the site-mean directions of the high temperature component: Lower (upper) hemisphere directions are represented by solid (open) symbols. (a) Sharilyn Fm (~155 Ma) and (b) Tsagantsav Fm (~130 Ma) in southern Mongolia.

removal of the LTC, a stable HTC decayed toward the origin near the unblocking temperatures of 580 or 660 °C. The HTC was dual-polarity, with magnetization directed north-northeastward and moderately down as normal polarity and south-southwestward and moderately up as reversal polarity (Figure 10a).

All of the site-level VGPs of the Tuchengzi Fm sandstones are Fisher-distributed because the results of the quantile-quantile plot test (Fisher et al., 1987) are positive (Mu = 1.127 < 1.207, Me = 0.864 < 1.094). Furthermore, the E/I method (Tauxe & Kent, 2004) was used for all 107 sandstone samples. The f value of the flattening correction factor was 0.9. The inclination was corrected from 59.9° to 62.5°, with the 95% confidence interval between 59.8° and 74.8°. These observations suggest that the results from the sandstones of the Tuchengzi Fm had not been significantly affected by inclination shallowing. Therefore, we use the HTCs without an E/I correction for analysis and discussion.

The site-mean direction for 21 sites is Dg = 14.5°, Ig = 58.3°, k = 39.2, and α_{95} = 5.1° in situ, which is significantly different from the modern field direction, and Ds = 10.1°, Is = 60.2°, k = 103.4, and α_{95} = 3.1° after tilt correction (Figure 10a and Table 1). The HTCs pass the fold tests of both McElhinny (1964) and McFadden (1990) at the 95% and 99% confidence levels (Table 1), and the dual polarity directions pass a reversal test

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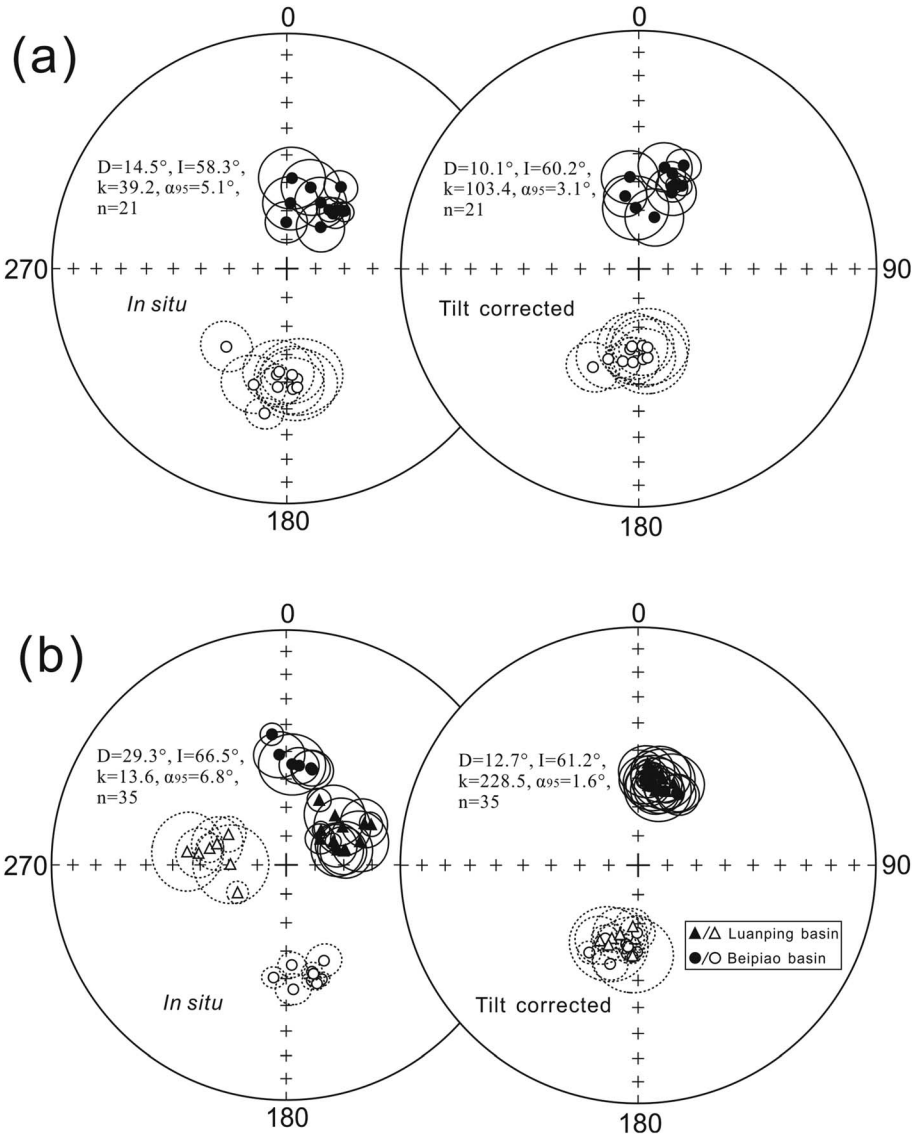


Figure 10. Equal-area stereographic projections of the site-mean directions of the high temperature component: Lower (upper) hemisphere directions are represented by solid (open) symbols. (a) Tuchenzi Fm (~155 Ma) and (b) Dadianzi and Yixian Fms (~130 Ma) in the Luanping basin and Beipiao basin of the North China Block, respectively.

(McFadden & McElhinny, 1990) at the 95% confidence level (class B) with an angle between the two averages γ_o = 4.6 < $\gamma_{critical}$ = 6.2. Thus, the HTC of the Tuchengzi Fm sandstones represents a prefolding primary magnetization.

Therefore, we averaged the site-level VGPs of 21 sites from Member 3 of the Tuchengzi Fm to obtain a ~140 Ma mean pole at 82.7°N/208.6°E (A_{95} = 4.3°) for the NCB (Table 1).

4.2.2. The Lower Cretaceous (~130 Ma) Dadianzi and Yixian Fms

In the Luanping Basin, 119 specimens were collected from the top of the Dadianzi Fm (~130 Ma) has from 20 sites in two sections and were demagnetized, including 52 specimens of volcanic rocks from 9 sites and 67 specimens of sedimentary rocks from 11 sites (Table 1). All specimens from these 20 sites yielded a well-defined LTC direction, the in situ coordinates of which cluster around the local modern geomagnetic field direction (D = −7.2°, *I* = 59.8°, IGRF online data, Figure 8d), representing a VRM of the present magnetic field. After removal of the LTC, a stable HTC of the volcanic and sedimentary rocks decayed toward the origin at an unblocking temperature of 580 °C. The representative Zijderveld plots are shown in Figures 7i and 7m. Dual

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polarity directions were observed from the HTCs (Figure 10b). All of the HTCs pass the fold tests of both McElhinny (1964) and McFadden (1990) at the 95% and 99% confidence levels (Table 1), which strongly suggests that the HTC was acquired before tilting. The dual polarity directions pass a reversal test (McFadden & McElhinny, 1990) at the 95% confidence level (class A) with an angle between the two averages $\gamma_o = 4.0 = \gamma_{critical} = 4.0$. Thus, the HTC is interpreted as a primary magnetization. All site-level VGPs of the [1]volcanic and sedimentary rocks of the Dadianzi Fm are Fisher-distributed because the results of the quantile-quantile plot test (Fisher et al., 1987) are positive (Mu = 0.928 $<$ 1.207, Me = 0.561 $<$ 1.094), and the HTCs of the volcanic rocks and sedimentary rocks are consistent. Thus, the results from the volcanic and sedimentary rocks are free of PSV influence and inclination shallowing, respectively. The site-mean HTC for 20 sites is Dg = 74.2°, Ig = 68.0°, k = 55.4, and α_{95} = 4.4° in situ, and Ds = 12.0°, Is = 61.2°, k = 263.4, and α_{95} = 2.0° after tilt correction (Figure 10b).

In the Beipiao Basin, 122 core samples were collected from basalt flows at 15 sites in two sections at the bottom of the Yixian Fm (Table 1). The NRM intensities ranged from 231 to 1,050 mA/m; the representative Zijderveld plots are shown in Figures 7n–7f. The basalt specimens from eight sites (four sites in the Sihetun section and four sites in the Huangbanjigou section) yielded a well-defined LTC direction, the in situ coordinates of which cluster around the local modern geomagnetic field direction (D = −8.3°, *I* = 60.1°, IGRF online data, Figure 8d), representing a VRM of the recent magnetic field. In most cases, a stable HTC decayed toward the origin at unblocking temperatures of 580 or 680 °C. Only one polarity was observed in each section. The HTC of the Sihetun section is directed southwestward and up, and that of the Huangbanjigou section is directed northeastward and down (Table 1). All the HTCs of the two sections pass the fold test of McFadden (1990) at the 99% confidence level and a reversal test (McFadden & McElhinny, 1990) at the 95% confidence level (class B) with an angle between the two averages $\gamma_o = 3.1 < \gamma_{critical} = 5.8$, which strongly suggests that the HTC of the Yixian Fm basaltic lavas is a prefolding primary magnetization. The site-mean direction for 15 sites is Dg = 355.3°, Ig = 51.3°, k = 100.1, and α_{95} = 3.8° in situ, and Ds = 13.6°, Is = 61.0°, k = 184.7, and α_{95} = 2.8° after tilt correction (Table 1). For the PSV test, all the site-level VGPs of the Yixian Fm basalts are Fisher-distributed because the results of the quantile-quantile plot test (Fisher et al., 1987) are positive (Mu = 1.053 $<$ 1.207, Me = 0.522 $<$ 1.094). The A_{95} value, which is obtained from the VGPs of 116 lava specimens, is 1.9, which is consistent with an N-dependent A_{95} envelope with a 95% confidence interval (1.79, 4.10), as proposed by Deenen et al. (2011, 2014). Moreover, the HTCs of the Yixian Fm were identical with those of the Dadianzi Fm (Figure 10b and Table 1). All of these observations indicate that the volcanic rocks of the bottom of the Yixian Formation had not been affected by the PSV.

Therefore, the site-level VGPs of 35 sites from the top of the Dadianzi Fm and the bottom of the Yixian Fm are averaged to obtain a ~130 Ma pole at 80.5°N/197.4°E (A_{95} = 2.3°) for the NCB (Table 1).

5. Discussion

5.1. Analysis of Paleopoles of the Major Blocks Surrounding the AMU-NCB

In this paper, we applied the widely accepted seven-point data quality criterion system (Van der Voo, 1990) to appraise the paleomagnetic data, and compiled the results in Table 2 with Q $\geq$ 4. Significantly, the reliable paleomagnetic data from volcanic rocks must average out the PSV, and those from sedimentary rocks have not been affected by inclination shallowing.

5.1.1. AMU and NCB

Currently, Late Jurassic–Early Cretaceous paleomagnetic data from the AMU remain scarce (Table 2 and Figures 11a–11c) because of poor age constraints or remagnetization. The Middle-Late Jurassic paleomagnetic pole from the Far East of Russia are suspected to have been subjected to local vertical-axis rotations (Kravchinsky, Sorokin, et al., 2002). And the Late Jurassic paleomagnetic poles from Inner Mongolia have a wider age range from ~139 to ~199 Ma (Zhao et al., 1990), and those from the Tergen and Shadaron Fms of the AMU underwent a remagnetization event in the Early Cretaceous (Cogné et al., 2005; Kravchinsky, Cogné, et al., 2002). In addition, some Early Cretaceous paleomagnetic data that were sampled from the southern MOS zone incorporate suspected local vertical-axis rotations (Halim et al., 1998) or late remagnetization (Cogné et al., 2005). Kovalenko (2010) and Shcherbakova et al. (2011) have carried out the paleomagnetic study for the ~130 and ~110 Ma volcanic rocks from southern Mongolia, but there are considerable contradictions within their results, which were likely caused by a small number of samples. Thus, we have

Table 2
The Mesozoic Paleopoles From the Siberia, Europe, Amuria, and North China Blocks

Age (Ma)	N	Plat (N°)	Plong (E°)	A_{95} (°)	Criterion (Q)	References
Siberia Block (SIB)						
82–74	16	82.8	188.5	6.1	123C5R7	Metelkin, Gordienko, et al. (2007)
130–110 [a]	25	72.3	186.4	6.0	123F5R7	Metelkin et al. (2004)
130–112 [b]	24 s	77.1	166.5	11.2	1–3–5-7	Kravchinsky (1995)
120 [(2)] mean [a-b]	26	72.5	189.7	5.8		Site-level VGPs' mean
140–120	11	67.2	183.8	7.8	123F5–7	Metelkin et al. (2008)
136–133	31 s	73.0	178.0	5.5	123–5-7	Pospelova (1971)
140–138	3	63.0	174.0	5.5	123–5-7	Pospelova et al. (1968)
146–136 [(3)]	4	62.3	168.6	11.8	1–3–5-7	Kravchinsky (1995)
~145	209 s	76.9	179.3	4.8	123–5-7	Houša et al. (2007)
155 [(4)]	18	63.6	166.8	8.5	123F5R7	Metelkin, Kazansky, et al. (2007)
155 [(1)]	9	64.4	161.0	7.0	123F5–7	Kravchinsky, Cogné, et al. (2002)
170–160	4	59.3	139.2	5.7	123F5–7	Metelkin et al. (2008)
J_1 (201–174)	5	43.3	131.4	23.2	1–3–5-7	Cogné et al. (2005)
245–176	26 s	47.0	129.0	8.0	123–5-7	Pisarevsky (1982)
North China Block (NCB)						
K_{1-2} (~100)	**10**	**74.5**	**201.0**	**4.7**	**123F5–7**	**Gilder and Courtillot** (1997)
114–110 [(7)]	**31**	**80.3**	**200.3**	**3.2**	**123–5-7**	Zhu et al. (2008)
121 [c]	12	82.8	160.5	9.4	123–5-7	He et al. (2008)
123 [d]	4	87.6	218.2	4.6	123–5-7	Zhu et al. (2004)
119 [e]	7	83.8	222.5	3.6	123–5-7	Zhu et al. (2002)
122–120 [f]	20	86.1	205.9	2.3	123–5-7	Zhu et al. (2001)
K_1 (145–100) [g]	10	75.8	208.7	7.4	123–5R7	X. Ma et al. (1993)
120 [(9)] mean [c-g]	**5S**	**83.7**	**202.0**	**5.1**		
130 [(8)]	**35**	**80.5**	**197.4**	**2.3**	**123F5R7**	**This study**
140 [(8)]	**21**	**82.7**	**208.6**	**4.3**	**123F5R7**	**This study**
155	*9*	*59.9*	*240.3*	*6.8*	*123–5R7*	Pei et al. (2011)
155 [(8)]	**35**	**69.6**	**203.0**	**5.6**	**123F5R7**	Ren et al. (2016)
J_3	10	74.4	222.8	5.9	123F5R7	Gilder and Courtillot (1997)
J_3	5	77.8	235.9	6.9	123F-R7	Gilder et al. (1999)
J_3	6	73.5	207.8	5.7	123F-R7	Gilder et al. (1999)
J_2 (~169)	**23**	**73.6**	**249.3**	**4.9**	**123F5R7**	Gilder and Courtillot (1997)
J_1 (~188)	**10**	**82.4**	**286.0**	**6.8**	**123–5R7**	Z. Y. Yang et al. (1998)
T_3 (~220)	**11**	**62.3**	**7.7**	**3.8**	**123F5R7**	Z. Y. Yang et al. (1998)
Amuria Block (AMU)						
~95 [(6)]	**23**	**84.7**	**195.0**	**6.6**	**123–5-7**	Hankard et al. (2007)
118–98 (~105) [(6)]	**27**	**80.5**	**159.0**	**5.7**	**123–5-7**	Hankard et al. (2007)
125–95 (~110) [(6)]	**126**	**80.8**	**158.4**	**2.5**	**123–5-7**	van Hinsbergen et al. (2008)
120 (?) [(13)]	*12*	*70.8*	*322.4*	*5.2*	*123–5 - -*	Cogné et al. (2005)
125–133 [(13)]	*12*	*86.8*	*61.8*	*7.3*	*123–5 - -*	Cogné et al. (2005)
130 [(5)]	**28**	**74.6**	**194.7**	**2.9**	**123F5R7**	**This study**
K_1	*14 s*	*58.3*	*51.0*	*4.2*	*1–3–5-7*	Halim et al. (1998)
155 [(5)]	**16**	**74.7**	**232.5**	**3.7**	**123F5R7**	**This study**
J_{2-3} [(13)]	*6*	*73.3*	*275.9*	*6.3*	*123–5 - -*	Cogné et al. (2005)
J_{2-3} [(12)]	*8*	*68.6*	*261.8*	*4.1*	*123–5 - -*	Kravchinsky, Cogné, et al. (2002)
J_{2-3} (178–157) [(14)]	*7*	*46.0*	*37.9*	*11.2*	*123F5–7*	Kravchinsky, Sorokin, et al. (2002)
J (139–199) [(11)]	*8*	*73.0*	*254.8*	*7.8*	*- 23–5R7*	Zhao et al., 1990
J (139–199) [(10)]	*14*	*62.4*	*224.6*	*4.9*	*- 23F5R7*	Zhao et al., 1990
Europe (EUR)						
100	12	81.7	180.1	6.7		Besse and Courtillot (2002)
130	14	75.8	192.9	2.8		Besse and Courtillot (2002)
140	7S	73.8	197.6	6.0		Besse and Courtillot (2002)
155	7S	74.3	137.4	3.5		Besse and Courtillot (2002)
190	18S	65.3	98.4	5.6		Besse and Courtillot (2002)
Tarim (TAR)						
K_1	13	64.1	172.1	12.0	123F5R7	Gilder et al. (2003)
J_3	6	64.6	208.9	9.0	123–5R7	Li et al. (1988)
J_1 (190 Ma)	2S	52.9	160.5	-		Huang et al. (2018)

Note. K_{1-2} = Early-Late Cretaceous; J_2 = Middle Jurassic; J_1 = Early Jurassic; N = number of sites samples (s) or Studies (S) used for pole determination; Plat and Plon = latitude and longitude of the pole; A_{95} = the radius that the mean pole lies within 95% confidence; Criterion (Q, numbers of the criteria met), reliability criteria 1–7 from Van der Voo (1990) [1 = well-dated rocks; 2 = sufficient numbers of samples, $n > 25$, $k < 10$ and $\alpha_{95} < 16°$; 3, details of demagnetization; 4 = field test (F, the positive fold test; C, positive baked-contact test); 5 = tectonic coherence with the craton and structural control; 6 = presence of reversals; 7 = no similarity to younger paleopoles in the same craton]; "-" = failed to meet this criterion; [a-g] = used for averaging the poles. [(1)–(14)] = corresponding to paleomagnetic sampling site number in Figure 1. Italicized entries were not used in the calculation of the apparent polar wander path. For the North China Block and Amuria Block, bold entries are used to refine the apparent polar wander paths. The reconstructions of Kazakhstan in Figure 13 used the poles of Europe.

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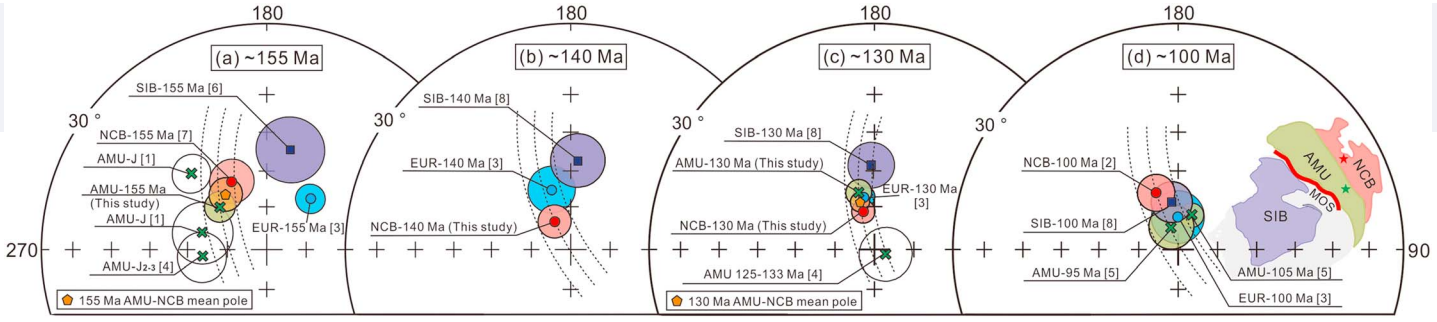


Figure 11. Equal-area projection (>30°N region only) showing the comparison of the coeval paleopoles of NCB, AMU, SIB, and EUR at the ~155, ~140, ~130, and ~100 Ma. The green star in AMU is the reference point (44.5°N, 109.3°E) for calculating the small-circle with its 95% errors in (a) and (c), and the red star in NCB is the reference point (41.6°N, 120.8°E) for calculating the small-circle with its 95% errors in (b) and (d). The poles with no color circle of A_{95} are problematic. The references for paleopoles: [1] Zhao et al. (1990), [2] Gilder and Courtillot (1997), [3] Besse and Courtillot (2002), [4] Cogné et al. (2005), [5] Hankard et al. (2007), [6] Metelkin et al. (2007b), [7] Ren et al. (2016) (the detailed data see Table 2), and [8] running mean poles for Siberia (see Table 3). NCB = North China Block; AMU = Amuria Block; SIB = Siberia; EUR = Europe.

omitted these problematic Late Jurassic–Early Cretaceous poles of the AMU. The Late Mesozoic available poles (Q ≥ 4) for the AMU were only obtained from the late Early Cretaceous volcanic rocks on the southern Mongolia (Table 2; Hankard et al., 2007; van Hinsbergen et al., 2008). Our new ~155 Ma Sharilyn Fm and ~130 Ma Tsagantsav Fm paleopoles from southern Mongolia, which satisfied all the seven-point data quality criterion (Van der Voo, 1990), can represent the Late Jurassic and Early Cretaceous key poles, and thus refined a segment of Mesozoic APWPs for the AMU (Figure 12).

For the NCB, we accept some reliable data from the Late Jurassic–Early Cretaceous that were reviewed by Van der Voo et al. (2015) and Huang et al. (2018). The high-quality paleopoles from the well-dated (~155 Ma) Tiaojishan Fm volcanic rocks in the Yanshan belt (Ren et al., 2016), for which the PSV has been averaged out, are consistent with the coeval poles from the red beds in the southern NCB (Gilder et al., 1999; Gilder & Courtillot, 1997). Additional reliable paleopoles from the middle–upper Yixian Fm (~120 Ma) basaltic lavas in west Liaoning (He et al., 2008; Zhu et al., 2001, 2002, 2004), which are also free of PSV, are in good agreement with the pole by X. Ma et al. (1993) obtained from the red beds of the Early Cretaceous Zhidan Group (~145–100 Ma) in the Ordos Basin of the central NCB. These reports all suggest that the paleomagnetic results from the Late Jurassic–Early Cretaceous clastic rocks of the NCB were free of inclination shallowing. Therefore, we averaged all of the ~120 Ma poles to obtain a mean pole (Table 2). Because the existing paleomagnetic database of the NCB lacks a reliable pole from ~155 to ~120 Ma, our new ~130 and ~140 Ma key poles for the NCB, which satisfied all data quality criteria (Van der Voo, 1990) and were free of PSV and inclination shallowing, can fill this vacancy. By combining our new key paleopoles with the published high-quality poles (Table 2), we constructed a new segment of Mesozoic APWP for the NCB (Figure 12).

Recently, several papers reviewed the Mesozoic paleomagnetic data of the NCB and calculated the APWPs (e.g., Huang et al., 2018; Van der Voo et al., 2015; Wu, Kravchinsky, & Potter, 2017). To better compare our new APWP with their results, we have plotted the Jurassic to Early Cretaceous segments of these APWPs of the NCB in Figure S1. The published APWPs include two different versions. Version 1 was constructed using the paleomagnetic data with inclination shallowing correction (Figure S1; Van der Voo et al., 2015; Wu, Kravchinsky, & Potter, 2017-version 1) and vision 2 using the data without inclination shallowing correction (Figure S1; Wu, Kravchinsky, & Potter, 2017-version 2; Huang et al., 2018). Our new APWP has a significant discrepancy from that of version 1 (Figure S1). That is because the version 1 used an f = 0.6 to "correct" all the data of clastic rocks. We do not agree that because we find that the Late Jurassic–Early Cretaceous paleopoles from volcanic rocks are consistent with the coeval poles from clastic rocks, indicating that there is no significant inclination shallowing in north China. Our new APWP revised the version 2 based on a new selection and adding some new data.

5.1.2. Siberia and Europe

Previously, many researchers considered Siberia to have been integrated with the European plate since ~250 Ma (Besse & Courtillot, 2002; Pavlov et al., 2007; Torsvik et al., 2012; Van der Voo et al., 2015), and thus a synthetic APWP during the Mesozoic was constructed for Europe–Siberia (Besse & Courtillot, 2002;

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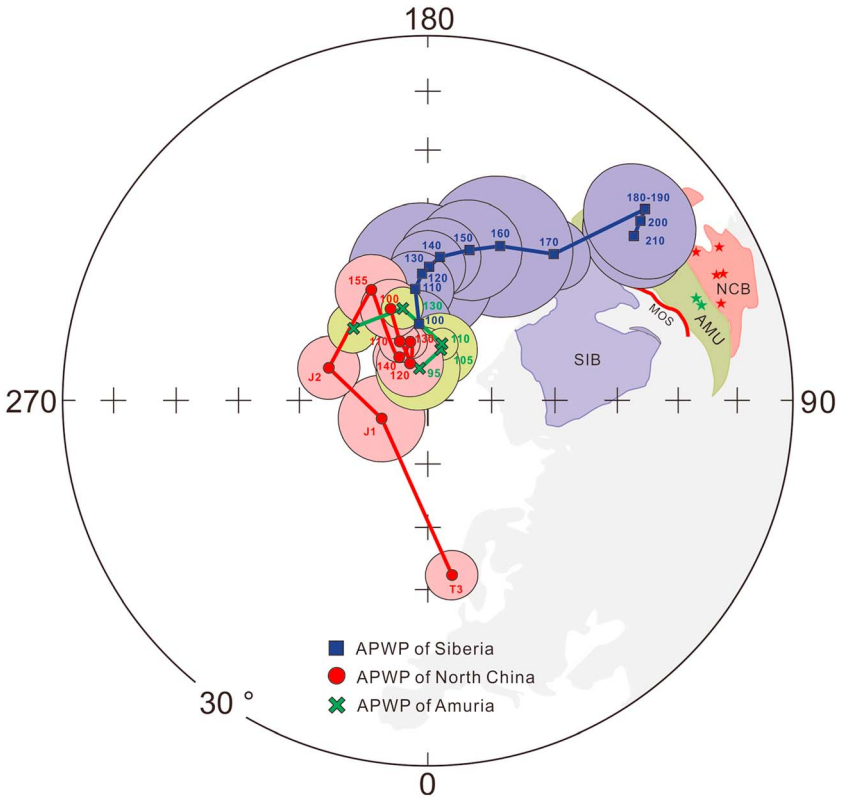


Figure 12. The new APWPs for the SIB, NCB, and AMU in the equal-area projection (>30°N region only). The stars are the paleomagnetic sampling locations in the NCB and AMU. The 180–190 and 210 Ma poles of Siberia are shown without theirs 95% errors because of $A_{95} > 16°$. APWP = apparent polar wander path; SIB = Siberia; NCB = North China Block; AMU = Amuria Block; MOS = Mongol-Okhotsk suture.

Schettino & Scotese, 2005; Torsvik et al., 2012). However, Metelkin et al. (2010) reported some new Mesozoic key poles for Siberia, which strongly demonstrated that there was relative tectonic rotation between Siberia and Europe along a large-scale strike-slip tectonic system during the Late Mesozoic. The geological observations also suggested that there was a multiple stage evolution for the strike-slip system during the Mesozoic time, for example, the large right-lateral strike-slip fault between Siberia and Kazakhstan since the Triassic period was observed by Kuzmin et al. (2010); Tomurtogoo et al. (2005) reported that there was a ~400 m wide Moron left-lateral shear zone on the west segment of the MOS zone during Late Jurassic, and they obtained the U–Pb zircon age of 172 Ma for mylonite sample from Muron river; moreover, the large-scale deformation (such as contorted and fragmented structures) within the western Central Asia Orogenic belt (Bazhenov & Mossakovsky, 1986; Gilder et al., 2008; Metelkin et al., 2010; Natal'in & Sengör, 2005). Thus, we must further separate the APWPs of Siberia and Europe, as discussed below.

For Siberia, Pavlov (2012) and Didenko (2015) reviewed the Mesozoic paleomagnetic data, but they excluded the data from the Transbaikalia where was suspected of having experienced local-axis rotation. However, two ~155 Ma high-quality paleopoles of Kravchinsky, Cogné, et al. (2002) and Metelkin, Gordienko, et al. (2007) from different locations in the Transbaikalia region were consistent, suggesting that these areas had kept the tectonically coherence since that time. Additionally, Metelkin et al. (2010) reported that the Late Mesozoic paleopoles from Transbaikalia coincide with the smoothed path of Siberia craton, we thus considered that Transbaikalia and Siberia craton likely formed a coherent tectonic domain since Late Jurassic.

In this paper, we collected all the paleomagnetic data available from Siberia and list them in Table S1. These data were obtained from Siberia craton, Transbaikalia, Minusinsky basin, Yenisei-Khatanga basin where have merged as a single unit since Late Mesozoic (Gordienko & Kuz'min, 1999; Metelkin et al., 2004, 2010; Nikishin et al., 2010). To more explicitly compare the Siberian paleopoles, we plotted all of them in Figure S2. Nevertheless, the data distribution is rather scattered. The main reasons are as follows:

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Table 3
Recalculated Apparent Polar Wander Path for Siberia

Age (Ma)	No. poles	Plat (N°)	Plong (E°)	A_{95} (°)
100	-	77.8	187.4	5.2
110	1	72.5	189.7	5.8
120	2	69.9	186.4	12.4
130	4	69.0	180.8	6.1
140	5	68.6	176.0	6.4
150	4	66.9	167.5	8.3
160	3	63.0	154.9	11.3
170	1	59.3	139.2	5.7
180	1	43.3	131.4	16.0
190	1	43.3	131.4	16.0
200	2	45.2	130.2	8.9
210	1	47.0	129.0	8.0

Note. Apparent polar wander path was recalculated from the Table 3 using a running mean at 10 Myr interval with a sliding window of 20 Myr. The 100 Ma pole is cited from Metelkin et al. (2012), and "-" means that the pole was calculated by running mean. Plat and Plon = latitude and longitude of mean pole; A_{95} = the radius that the mean pole lies within 95% confidence.

1. Some poles from Siberia cratonic margin, which were suspected of having experienced local-axis rotation (e.g., Davydov & Kravchinsky, 1971; Pospelova, 1971), were significantly difference from the poles in proper Siberia craton (Table S1 and Figure S2). For example, the Late Triassic–Early Cretaceous paleomagnetic poles from Chulym-Yenisey Depression (Table S1; e.g., Pole 7 from Pospelova 1971) and the Late Jurassic poles from Khatanga basin (Table S1; Pole 19 from Pospelova 1971) differ from the coeval poles of Siberia craton, suggesting that these basins experienced the suspected local rotation during that time.
2. Some results unpublished (e.g., Pole 8 from Global Paleomagnetic Data Base-2599 of Pisarevsky, 2005, and some poles from Kuhar, 1991; Pavlov & Karetnikov, 2008) or from meeting abstracts have no detailed demagnetization information (e.g., poles from Gusev, 1973; Pavlov, 2012; Rodionov & Osipova, 1979; Rozinov & Sholpo, 1971; Slautsitais, 1971).
3. Pole 11 in Table S1 (Pavlov et al., 2004) was obtained from igneous rocks being a virtual geomagnetic pole, which cannot average out PSV. We thus ruled out these problematic poles in the calculation of the APWP.

Here we applied the widely accepted seven-point data quality criterion system (Van der Voo, 1990) to appraise all the paleomagnetic data, and selected the results in Table 2 with $Q \geq 4$. We recalculated these data (Table 2) using a running mean with a 20 Myr sliding window every 10 Myr to obtain a new APWP for Siberia during 210–100 Ma (Table 3 and Figure 12).

For Europe, although the APWPs of Torsvik et al. (2012) have been used for discussion (e.g., Van der Voo et al., 2015; Wu, Kravchinsky, Gu, et al., 2017), they included Cretaceous data from Mongolia that are unfit to compare with our new Early Cretaceous AMU and NCB paleomagnetic results. Therefore, we continue to choose the frequently-used Europe APWP of Besse and Courtillot (2002), which was cited by Metelkin et al. (2010) and Ren et al. (2016).

5.1.3. Comparison of the Late Jurassic–Early Cretaceous Paleopoles

To better analyze and compare the APWPs of Siberia, the AMU, and the NCB, we translated these APWPs into relational kinematic parameter graphs of paleolatitude (Figure 13a) and paleoazimuth (Figures 13b and 13c) at a reference point (51.0°N, 112.0°E) on the MOS.

Our new ~155 Ma paleopole of the Sharilyn Fm from the AMU is nearly consistent with the coeval pole from the Tiaojishan Fm of the NCB (Ren et al., 2016; Figure 11a), which indicates that these two blocks had merged into a tectonically coherent unit (AMU-NCB) in the Late Jurassic. Thus, we averaged all site-level VGPs of the Sharilyn Fm and Tiaojishan Fm to obtain a ~155 Ma mean pole at 71.7°N/211.4°E (A_{95} = 4.1°) for the AMU-NCB. This ~155 Ma pole of the AMU-NCB is significantly different from the coeval poles of Siberia and Europe (Kravchinsky, Cogné, et al., 2002; Metelkin, Kazansky, et al., 2007; Figure 11a). For the reference point (51°N, 112°E) on the MOS, this difference of the paleopoles of the AMU-NCB and Siberia indicates that there was a latitudinal difference of 14.3° ± 6.9° and relative rotation of ca. 19.0° between Siberia and the AMU-NCB after ~155 Ma (Figures 13a and 13c).

Our new ~140 Ma Tuchengzi Fm paleopole for the NCB is different from the coeval pole of Europe, as shown by ca. 12.0° ± 7.2° relative rotation along the small circle (Figure 11b) for the reference point (41°N, 117°E). In addition, the ~140 Ma pole of the NCB has a significant discrepancy from the coeval pole of Siberia (Figure 11b), as indicated by the presence of the differences of the ca. 6.0° paleolatitude (Figure 13a) and ca. 24.0° paleoazimuth (Figure 13c) between them. Our new ~130 Ma poles of the AMU and NCB overlap with each other, which further indicates that the AMU-NCB remained as a stable unit during that time. Therefore, we averaged them to obtain a ~130 Ma mean pole at 77.9°N/195.9°E (A_{95} = 1.9°) for the AMU-NCB. It is indistinguishable from the coeval pole of Europe, which indicates that the AMU-NCB had been merged as a unit with the Europe by ~130 Ma (Figure 11c). However, the ~130 Ma pole of the AMU-NCB is still somewhat different from the coeval pole of Siberia (Figure 11c). Although their paleolatitudes were identical (Figure 13a), ca. 19.4° relative tectonic rotation between Siberia and NCB occurred after ~130 Ma

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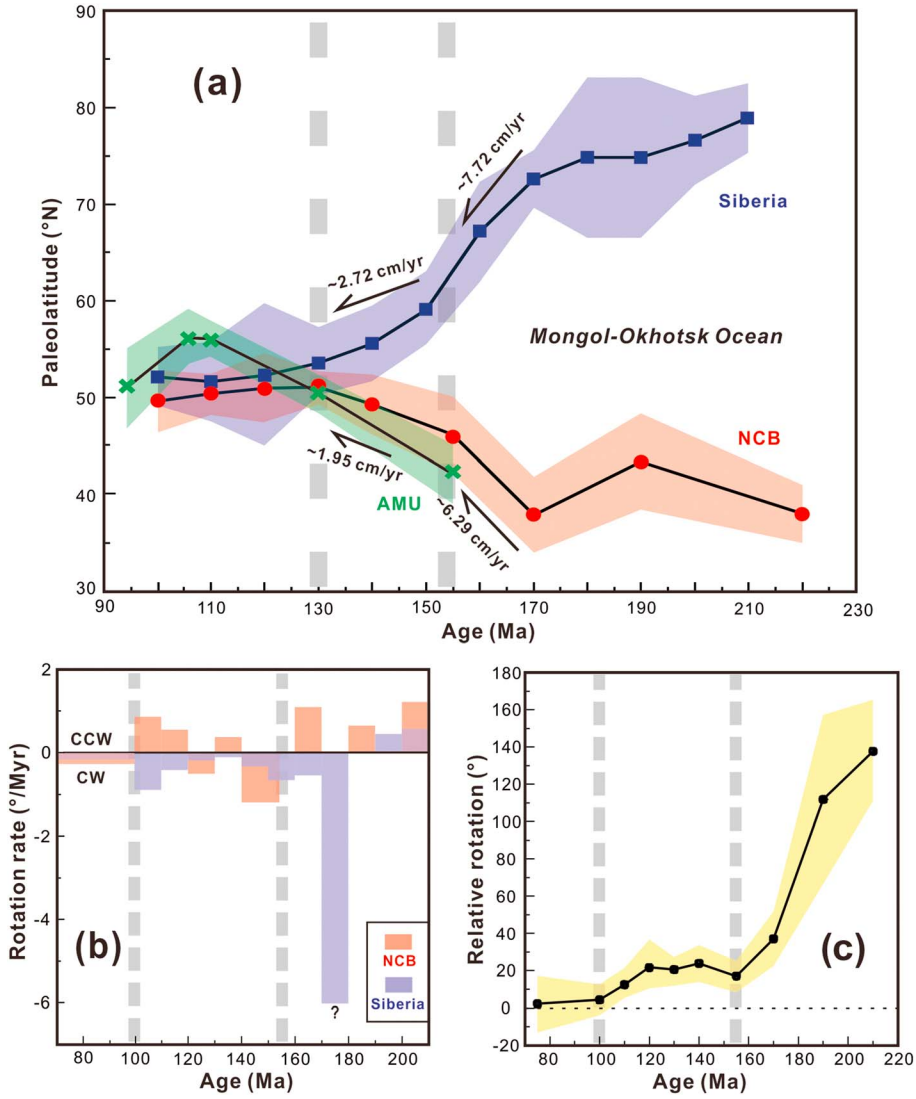


Figure 13. The parameter curves versus age showing in the (a) paleolatitude for the Siberia, NCB and AMU, (b) rotation rates for Siberia and NCB, and c relative rotation between Siberia and NCB. These parameters are calculated by paleopoles from the apparent polar wander paths of Siberia and NCB at reference point (51°N, 112°E) on the Mongol-Okhotsk suture. The CW and CCW in (b) represent clockwise and counter-clockwise, respectively. The gray dashed lines represent the change point. NCB = North China Block; AMU = Amuria Block.

(Figure 13c). Up to the latest Early Cretaceous (~100 Ma), the paleopoles of the NCB and Siberia are indistinguishable (Figure 11d).

5.2. The Evolution of the MOS During the Jurassic to Early Cretaceous

To ascertain the kinematic characteristic of the AMU, NCB, and Siberia and the subduction-collision process of the evolution of the MOS, we propose a new evolutionary model based on the current paleomagnetic and geological information (Figure 14).

In this paper, we do not discuss the models and timing for the opening of the Mongol-Okhotsk Ocean because this topic is still under debate (Donskaya et al., 2013; Kuzmin & Fillipova, 1979; Tomurtogoo et al., 2005; Zorin, 1999). However, we accept that the Mongol-Okhotsk oceanic slab had begun to bidirectionally subduct by the Late Paleozoic because intense subduction-related magmatic activities were recorded on both sides of the ocean since the Carboniferous (Figure 1b; Donskaya et al., 2013; Sun et al., 2013; W. Wang et al., 2015). The progressive plate convergence of the two sides of the Mongol-Okhotsk Ocean continued during the Mesozoic (Figure 13).

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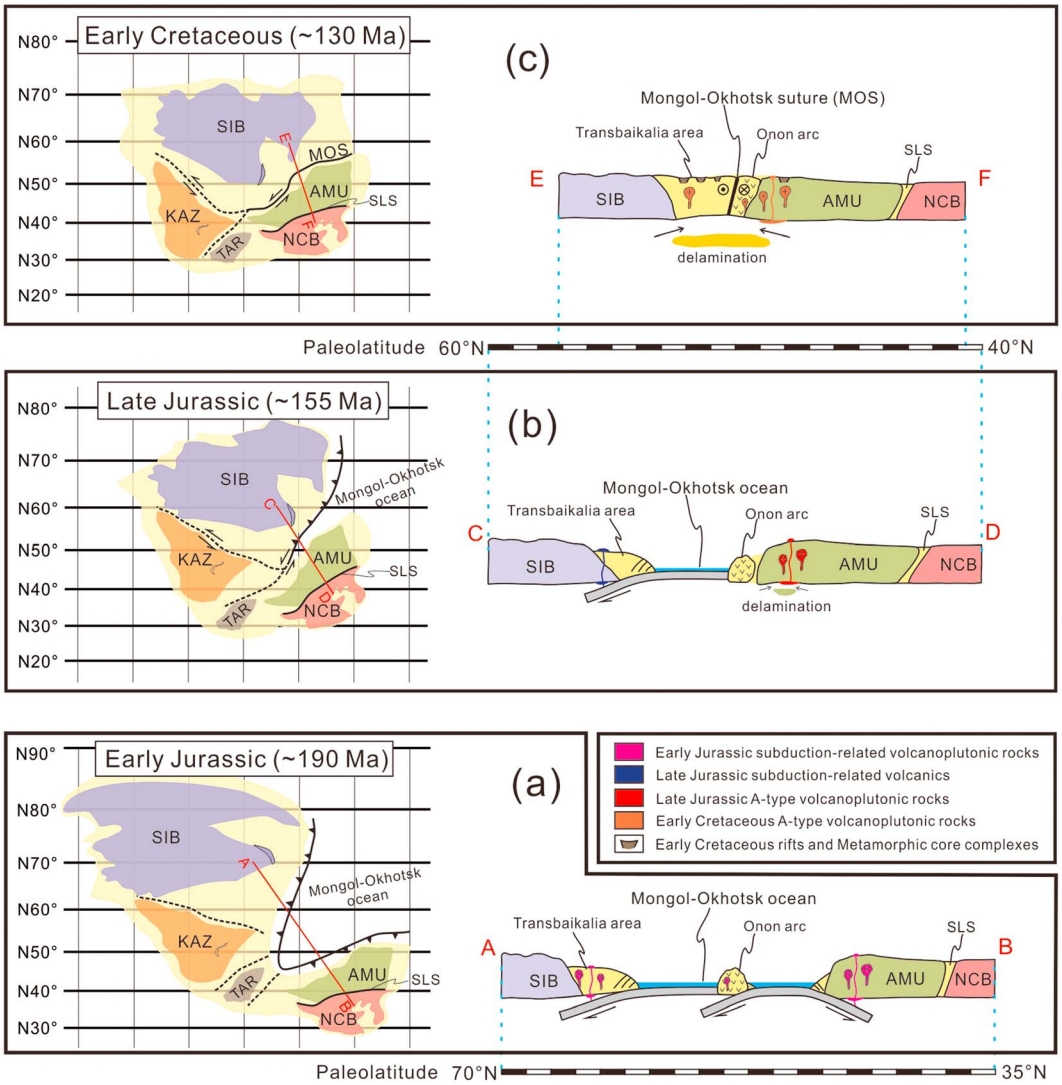


Figure 14. Schematic reconstruction of the SIB, AMU, and NCB from the Early Jurassic to Early Cretaceous using the Euler rotation parameters. Euler rotation parameters: Early Jurassic, SIB (32.6, 54.1, 43.5), KAZ (−41.4, 42.4, 29.3), NCB/AMU (81.5, −22.2, −35.7), TAR (−37.1, 59.9, 31.6); Late Jurassic, SIB (36.5, 86.8, 33.0), KAZ (0, 69.9, 15.0), NCB/AMU (0, 113.0, 20.4), TAR (7.7, 89.1, 26.5); and Early Cretaceous, SIB (8.0, 100.4, 17.7), KAZ (0, 102.9, 14.2), NCB/AMU (35.8, 293.8, −10.2), TAR (18.4, 97.1, 20.5). The model of the evolution of the Mongol-Okhotsk suture corresponds to the reconstructions. For abbreviations see caption of Figure 1.

During the Early Jurassic, the paleolatitude difference between Siberia and the NCB was ca. 32.5° (Figure 13a). However, the western part of the AMU-NCB had connected with Siberia and formed an enormous gulf in the Early Jurassic (Figure 14a), and its central width may have reached ca. 3,575 km between the AMU and Siberia. During this period, Siberia experienced rapid clockwise rotation (Figure 13b). This significant plate movement was accompanied by the bivergent subduction of the oceanic slab, which was reflected by the records of subduction-related magmatic activities on both sides of the Mongol-Okhotsk Ocean (Figures 1b and 14a; Donskaya et al., 2013; W. Wang et al., 2015).

After the Early Jurassic, Siberia and the AMU-NCB experienced rapid southward and northward convergence with rates of latitudinal movement of ~7.72 and ~6.29 cm/year, respectively, which then became slower at ~155 Ma (Figure 13a), as well as the variation of the relative rotation between them (Figure 13c). This important change at ~155 Ma may have resulted from two different dynamical mechanisms: (1) the effect force was abruptly weakened at the southern margin of the NCB in the Late Jurassic

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because the diachronous collision of the NCB and the South China Block had finished by the Middle–Late Jurassic, based on paleomagnetic investigations (e.g., Gilder & Courtillot, 1997; Pei et al., 2011; Uno & Huang, 2003; Van der Voo et al., 2015; Z. Yang et al., 1992; Z. Yang & Besse, 2001); and (2) the southward subduction of the Mongol-Okhotsk oceanic slab beneath the AMU-NCB ceased in the Late Jurassic. This second dynamical mechanism was explained by a special subduction model of K. J. Zhang (2014), which suggested that there was a northeast-trending mid-ocean ridge (MOI) from the Mongol-Okhotsk oceanic slab that subducted obliquely under the AMU-NCB in the Late Jurassic. This MOI would have been a long, narrow slab window that separated the oceanic slab into two parts. After the southeastern oceanic slab side of the MOI rapidly subducted underneath the AMU, the northwestern side would have abruptly lost drag force, which would have weakened or stopped its subduction. Thus far, geological investigations, which suggest no record of Late Jurassic subduction-related magmatic activity at the northern margin of the AMU, support the inference that the southward subduction of the Mongol-Okhotsk Ocean stopped at this time.

Our Late Jurassic (~155 Ma) reconstruction shows that there was massive paleolatitude difference of 14.3° ± 6.9° (ca. 1587 ± 766 km) between Siberia and the AMU-NCB, which suggests that the Mongol-Okhotsk Ocean had not yet closed completely by that time. This conclusion is supported by geological and geophysical evidence. For example, marine strata are exposed on the eastern MOS (e.g., the Mohe Basin; Guo et al., 2017; Zonenshain et al., 1990); some Late Jurassic subducted-related basaltic lavas in the Transbaikalia area (Figures 1b and 14b; Andryushchenko et al., 2010; Vorontsov et al., 2002) suggest that the northward subduction of the oceanic slab was continuing at that time; and seismic tomography analysis indicated that Late Jurassic Mongol-Okhotsk oceanic slab remnants remained beneath the Siberian margin (Van der Voo et al., 1999, 2015; Wu, Kravchinsky, Gu, et al., 2017). Although researchers have reported some Late Jurassic A-type volcanoplutonic rocks (with negative Eu anomalies and high alkaline contents) near the Manzhouli area (e.g., Sun et al., 2013; Tang et al., 2015; W. L. Xu et al., 2013), these magmatic rocks are only locally distributed near the eastern "Onon arc" zone (Figure 1b). And analysis of the geochemical characteristics indicates that the primary magma that formed the Late Jurassic A-type granitoid was probably derived from the partial melting of a delaminated region of the lower crust (Tang et al., 2015). Thus, we suggest that it represented a local extension setting characterized by the collapse or delamination of thickened continental crust after the arc-continent collision (Figure 14b).

The large-scale plate convergence after ~155 Ma was caused by the movement of both Siberia and the AMU-NCB. However, Siberia always maintained faster southward motion (Figure 13a) and steadier clockwise rotation (Figure 13b) than the AMU-NCB. Thus, Siberia played a major role in the convergence process. The ~1587 km latitudinal plate convergence between Siberia and the AMU-NCB probably ended by ~130 Ma (Figure 13a). This important event not only represents the complete closure of the Mongol-Okhotsk Ocean to form a northeast-trending suture, but also indicates that the intense collisional compression within the Mongol-Okhotsk tectonic domain was beginning to transform into post-collisional extension, which was represented by the widespread development of rift basins (Figure 14c; Meng et al., 2003; Metelkin et al., 2010; Y. T. Yang et al., 2015), post-orogenic A-type granitoids (Figures 1b and 14c; Sun et al., 2013; T. Wang et al., 2015) and metamorphic core complexes (Figures 1b and 14c; Daoudene et al., 2013, 2017; Donskaya et al., 2008; Meng, 2003; Sklyarov et al., 1994, 1997) on both sides of the MOS. The regional extension in the Mongol-Okhotsk tectonic domain was probably related to upwelling of asthenospheric material as a result of the post orogenic collapse or delamination of a thickened crust (Figure 14c), as suggested by the widespread nature of A-type volcanoplutonic rocks (Donskaya et al., 2013; Sun et al., 2013; Tang et al., 2015; T. Wang et al., 2015).

Although the latitudinal collision between Siberia and the AMU-NCB had ended by ~130 Ma, relative tectonic rotation, along with a large-scale Mongol-Okhotsk sinistral strike-slip system, had not yet stopped (Figure 13c). Siberia underwent clockwise rotation after ~130 Ma (Figure 13b), which may have been driven by the collision of the Kolyma-Omolon microcontinent with eastern Siberia (Metelkin et al., 2016; Oxman, 2003; Y. T. Yang et al., 2015). The AMU-NCB began to turn in a steady counter-clockwise rotation (Figure 13b) at ~120 Ma, which may have been driven by the mechanism of oblique subduction of the Pacific oceanic slab (Y. Wang et al., 2018; W. L. Xu et al., 2013). By ~100 Ma, the rotations of Siberia and the AMU-NCB had ceased (Figure 13b and c).

6. Conclusions

In this study, we report new paleomagnetic results from the ~155 Ma Sharilyn Fm and ~130 Ma Tsagantsav Fm of the AMU, and the ~140 Ma Tuchengzi Fm and ~130 Ma Dadianzi/Yixian Fm of the NCB. These data have not been affected by compaction-induced inclination shallowing or PSV, and satisfy the seven widely accepted quality criteria proposed by Van der Voo (1990). These new key poles refine a segment of Mesozoic APWPs for the AMU and NCB that overlapped with each other, which demonstrates that the AMU and NCB maintained tectonic coherence during that time. Those APWPs and the relative kinematic parameters of Siberia and the AMU-NCB indicate that there were 14.3° ± 6.9° (ca. 1,587 ± 766 km) latitudinal plate convergence and ca. 19.0° relative tectonic rotation between Siberia and the AMU-NCB after ~155 Ma. The large-scale latitudinal convergence likely ceased by ~130 Ma, although some relative rotation between them continued until ~100 Ma. This process was accompanied by a large-scale Mongol-Okhotsk sinistral strike-slip system between Siberia and the AMU-NCB.

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